

Gender Sensitive Neighborhood Park Planning in a Hot Desert City:
An Urban Development Initiative in Tempe, Arizona

Jordyn P. Hitzeman

School of Geographical Sciences and Urban Planning, Arizona State University

Honors Thesis

Dr. Meagan Ehlenz and Dr. Hue-Tam Jamme

February 11, 2025

Table of Contents

Table of Contents.....	1
Background.....	4
Literature Review.....	8
2.1 Introduction.....	8
2.2 Demographics and Neighborhood Park Use.....	8
2.2.1 The Gap: Neighborhood Park Use Disparities.....	8
2.2.2 Why don't teenage girls visit the park?.....	11
2.2.3 Practical Reasons for Motivating Park Use.....	14
2.2.4 Planning and Design Strategies for Increasing Usage.....	16
2.3 Reconciling Climate, Public Space Design, and Gender.....	18
2.3.1 Climate Considerations for Neighborhood Park Design.....	18
2.3.2 The Intersectionality of Climate and Gender.....	19
2.3.3 Gender Inclusion in Climate Discourse.....	21
Methodology.....	23
3.1 Research Questions.....	24
3.2 Process Overview.....	24
3.3 Sample Description.....	27
3.4 Survey Methodology.....	27
3.5 Workshop Methodology.....	28
3.5.1 Pre-Workshop.....	29
3.5.2 Workshop Introduction.....	29
3.5.3 Observations Template.....	30
3.5.4 Reflection.....	31
3.5.5 Solutions Discussion.....	32
3.5.6 Solutions Design Activity.....	32
Survey Results and Discussion.....	34
4.1 Evaluating Neighborhood Park Use.....	34
4.1.1 Visitation Frequency.....	34
4.1.2 The Nature of Adolescent Park Use.....	39
4.1.3 Summer Heat and Neighborhood Park Use.....	42
4.2 Motivating Neighborhood Park Visitation.....	46
4.2.1 What's Working?.....	47
4.2.2 What would the TGM like to change?.....	50

4.3 Survey Discussion.....	54
Workshop Results and Discussion.....	58
5.1 Observations Template.....	59
5.1.1 Design and Aesthetic.....	60
5.1.2 Comfort (Safety).....	63
5.1.3 Comfort (Maintenance and Environment).....	66
5.1.4 Activities.....	68
5.2 Designing Solutions.....	69
5.2.1 Design and Aesthetic Insights.....	69
5.2.2 Comfort (Safety) Requests.....	70
5.2.3 Comfort (Maintenance and Environment) Solutions.....	72
5.2.4 Activity Suggestions.....	73
Conclusions and Recommendations.....	79
6.1 Conclusions.....	79
6.2 Recommendations.....	85
6.2.1 Incorporate more native plants and other vegetation into park design.....	86
6.2.2 Bigger, better, shaded swings (and other active recreation amenities).....	87
6.2.3 Do not neglect the visual aspect of park features. Enhance them with public art....	89
6.2.4 Fix the bathrooms.....	92
6.2.5 Curate cozy spaces for socialization.....	92
6.2.6 Focus on nighttime safety. Lighting is a must, but be strategic.....	93
6.3 Limitations.....	94
6.4 Final Considerations.....	97
References.....	100
Appendices.....	109
Appendix A.....	114
Appendix B.....	115
Appendix C.....	125
Appendix D.....	131
Appendix E.....	137
Appendix F.....	138
Appendix G.....	141
Appendix H.....	143

Background

There is a growing chorus of urban voices calling for the inclusion of underrepresented perspectives in the urban development process. These voices emphasize how the needs and interests of adolescent girls and gender minorities, in particular, have been largely ignored in the planning and design of public spaces. As a result, adolescent girls and gender minorities are deterred from using these spaces and subsequently miss out on the social and health benefits associated with participating in them (Cohen et al., 2007; Bustamante et al., 2022). These voices also call for similar solutions: to build better spaces for adolescent girls and gender minorities, we should simply ask them what they want. More formally, this chorus advocates for the use of participatory action methodologies to address these urban design disparities.

Some voices speak to design more broadly. Caroline Criado-Perez's *Invisible Women* issues an edict of concern about the systemic lack of gender-disaggregated data used in research and design for the default male. From pianos to clinical trials to wood-burning stoves, the consequences of non-inclusive design are stark gender disparities when it comes to career, health, and general life enjoyment (Criado-Perez, 2019). Other voices focus on the urban form at large. In 2021, UN-Habitat and the Shared City Foundation launched *Her City*, a joint urban development initiative. The goal of *Her City* is to equip urban actors with the tools necessary to develop urban spaces from the girls' perspective. The *Her City Toolbox* includes "9 flexible building blocks" which outline a participatory action roadmap for stakeholder engagement,

citywide assessment, site-specific assessment, analysing challenges, designing ideas, and recommendations for action (Her City, 2021).

More targeted voices call for sweeping reforms to how urban actors approach the design and planning of public recreation spaces – specifically, public parks. Make Space for Girls, a Bristol-based charity, criticized the state of Global North public park design in a Bloomberg article appropriately titled “Teen Girls Need Better Public Spaces to Hang Out” (Lange, 2021). They lament that girls’ neighborhood park visitation drops off in their teenage years, most likely because default-adolescent amenities – such as basketball courts and skate parks – and abhorrent restroom conditions fail to meet their needs or serve their interests (Walker & Clark, 2023).

Another chorus is increasingly calling for new approaches to urban planning and design. Composed of urban academics, planners, and stakeholders residing in hot desert cities – urban areas that are especially vulnerable to rising global temperatures –, this chorus emphasizes the need for climate adaptation through design. A slew of recent research has been aimed at modifying the urban form to adapt to extreme heat and its potential impacts on urban life (Baker et al., 2002). With respect to the general urban landscape, proposals range from the implementation of vast urban tree canopies, fostering native plant growth, installing artificial shade structures, and opting for porous asphalt alternatives, which reflect more sunlight and absorb less heat (Baker et al., 2002; Midel et al., 2015). These same voices advocate for extending these heat mitigation efforts to public park design, while centering the inherent role that public parks play in mitigating extreme heat for the surrounding areas (Decelet-Barreto et al., 2013; Colter et al., 2019; Trego et al., 2023).

1. Background

Even though the targeted beneficiaries of gender sensitive design, women, stand to gain the most from climate sensitive design, these choruses rarely ever sing in unison. Given that women are disproportionately exposed to and affected by extreme heat, and that extreme heat poses an additional barrier to park use for young women who are already disinclined from visiting neighborhood parks (Mashhoodi, 2021; Greenfield et al., 2023), it is essential to understand the intersection between gender, age, and climate when it comes to understanding neighborhood park use. As far as I am aware, none of the Global North-based participatory action initiatives that seek to promote park use among adolescent girls and gender minorities have been executed in the unique geography of hot desert cities (Her City, n.d.). Further, the existing research efforts aimed at promoting equitable design for heat mitigation in hot desert cities rarely, if at all, account for adolescent girls and, even less often, gender minorities as a group worthy of special consideration (Hernandez et al., 2022).

The goal of this research is to gain a more workable understanding of how adolescent girls and gender minorities feel about and interact with their neighborhood parks, how their park use behaviors change from childhood to adolescence – specifically in hot desert cities – and how design-based interventions can promote park visitation among this demographic. This thesis employs a modified version of the methodology outlined by the *Her City Toolbox* to parse how adolescent girls and gender minorities in the Phoenix metro area perceive the quality of existing recreational public spaces that they have access to as well as define improvement from their perspective. Participants were recruited from the Girl Scouts–Arizona Cactus-Pine Council. They completed a survey detailing their relationships with neighborhood parks over time and answered specific questions about how heat affects their neighborhood park use. The participants then

1. Background

attended a workshop at a public park in Tempe, Arizona, where they worked in groups to assess the quality of the park from their perspective and worked individually to design a version of the park that met their unique needs and interests.

Adolescent girls and gender minorities have been designed out of public space, especially parks. Extreme heat can force everyone out of outdoor public spaces in hot desert cities. A design-based approach that incorporates the needs and interests of teenage girls while also prioritizing heat resilience is essential for re-engaging adolescent girls and gender minorities with public space year-round. The overworked city planner is left asking, “How?”

This research is not a question of whether or not these gender and age disparities exist as barriers to neighborhood park use, rather, it aims to unpack the extent to which these disparities exist at the intersection of Global North planning and hot desert climates. The majority of existing park use research that focuses on gender and age disparities is based in Northern Europe, where climates are not comparable to those of hot desert cities. Extreme heat stands to impose barriers to the neighborhood park visitation of adolescent girls and gender minorities, but it has yet to be academically explored. Not only does my research evaluate non-male adolescent park use in a major hot desert city, but it also directly investigates the relationship between extreme heat and park use within this demographic. My research consolidates insights from this demographic with existing design proposals for heat mitigation and gender inclusion to prescribe neighborhood park design solutions that are both climate and gender sensitive. Using a participant-centered methodology ensure that these strategies come straight from the source while also empowering adolescent girls and gender minorities with the knowledge to effectuate change in their own communities.

Literature Review

2.1 Introduction

The goal of my research is to reconcile two active discussions in the planning field tangential to public space design: demographic sensitivity and climate sensitivity. I specifically focus on the relationship between age, gender, and heat and how these factors affect neighborhood park use. Accordingly, the literature review will be divided into two distinct sections.

The first section will explore current understandings of how demographic characteristics shape individuals' engagement with public spaces and neighborhood parks. I will address gender and age disparities in neighborhood park and public space use, why they exist, why they are a problem, and what we have done and can do to address them.

In the second section, I will examine the ways in which climate motivations in hot desert metropolitan areas influence neighborhood park design. Finally, I will tie the conversation back to gender by exploring its critical intersection with climate and evaluating the relevance of this intersection in the context of neighborhood park planning and design.

2.2 Demographics and Neighborhood Park Use

2.2.1 The Gap: Neighborhood Park Use Disparities

2. Literature Review

Gender. A preponderance of evidence illustrates a significant and consistent disparity in park usage regarding gender, both in terms of visitation frequency and duration (Bornioli et al., 2024). One national sample found the gender breakdown of park users to be 59% men and 41% women (Evenson et al., 2019). Another study produced similar results, reporting that not only were 57% of total park users male, but men also accounted for 60% of the total hours spent in the neighborhood parks studied (Cohen et al., 2016). This disparity persists across multiple demographics including age, ethnicity, and socioeconomic status.

In Spain, ethnic disparities were apparent when analyzed through a gendered lens: the park use rates of Asian, Arab, and Latina women were considerably lower than their male counterparts. Further, among the women observed in the parks, it was more likely they were White or Latina than any other racial or ethnic group. The researchers suggest that this may be mediated by differences in gender roles between cultures as well as the enhanced vulnerability to physical violence and harassment that accompanies being both a woman and a minority (Perez-Tejera et al., 2018).

Regarding socioeconomic status, one research team found that in low-income Los Angeles neighborhoods, men not only outpaced women in both the number of self-reported park visits and the duration of such visits, but women were also significantly less likely to exercise in the park – or at all (Derose et al., 2015).

Age. A substantial body of research points to age as a predictive characteristic of park usage. Multiple studies have ranked park usage by age group, showing that adults account for the largest portion of park users, followed closely by children. However, there is a sharp drop off in

2. Literature Review

park use among teens and seniors (Evenson et al., 2019; Cohen et al., 2016; Marquet et al., 2019).

One study focusing on children's park use observed a critical underutilization of parks among their sample of 10- to 13-year-olds:

“Eight parks contained no children at all during our observation times, while another 10 parks contained no more than 10 children each.”

The study also noted that nearly twice as many boys used parks as girls within this specific age group, reflecting an exacerbation of gender-based park use disparities as children enter adolescence (Loukaitou-Sideris & Sideris, 2009).

Further research from a survey introduced earlier in this section emphasizes the pronounced gender disparities among teens,

“Male teens accounted for 65% of teen visitors and 68% of teen MVPA person hours ($p < 0.05$). Park visitation by age group was significantly different than that in the general population ($p < 0.0001$),” (Cohen et al., 2016).

Given the significant decline in park usage among girls as they enter adolescence, the remainder of the literature I review is focused on understanding the barriers that prevent teenage girls from using neighborhood parks as frequently as their peers. A diagnostic approach to these barriers necessitates a deep investigation into potential methods for addressing them.

Race and Ethnicity. While certain demographic characteristics are overwhelmingly predictive in determining an individual's degree of park usage, general racial and ethnic background does not appear to be one of them. Some evidence from a park use behavior study in Minneapolis suggests that Black Americans and those born outside of the United States tend to underutilize public parks (Das et al., 2017), but a Los Angeles-based analysis of racial-ethnic differences in park use found no significant variation in park visitation frequency along racial

lines. This contrast may be attributed to demographic differences in the neighborhoods studied and the subsequent social power dynamics that emerge. Nevertheless, race alone does not prove to be predictive in isolation. The literature does recognize differences in *how* specific racial-ethnic groups use community park spaces. For example, Black people and english-speaking Latinos did not use the parks for physical activity as much as White people, while spanish-speaking Latinos stood out for their use of parks as primarily social spaces (Derose et al., 2015). Some research focused on children's park use did find that hispanic children were more likely to use their neighborhood park than children of other racial groups, but, beyond that, race was more influential in determining the geography of children's park use (inner-city vs. suburban) than any other metric (Loukaitou-Sideris & Sideris, 2009).

2.2.2 Why don't teenage girls visit the park?

Safety, Intimidation, and Harassment. Perceived lack of safety, intimidation, and harassment are among the most common reasons women and gender minorities – especially teenage girls and gender minorities – express apprehension toward public space and neighborhood park use (Bornioli et al., 2024). Some academics argue that this is because the historical dominance of men in the planning field has resulted in women's issues being ignored on the planning agenda, safety being one of those issues (Listerborn, 2016). Safety concerns not only prevent teenage girls from wanting to use their neighborhood parks, but it causes reluctance among their parents over whether or not to allow them to use the park alone. A study investigating how different factors affect children's park use found that while most parents stated that they did not let their child use the park alone, the girls in the study had “*decidedly less independent mobility*.” In addition, among the middle school aged children surveyed in Los

2. Literature Review

Angeles living in both suburban and inner city areas, boys were significantly more likely to report feeling safe than girls were (Loukaitou-Sideris & Sideris, 2009).

A fundamental piece of literature written about women and public space, *The Geography of Women's Fear* speaks to how intimidation and harassment, or fears of such, contribute to women and girl's apprehension toward using public spaces, claiming that women learn early that it is not safe to be alone in male dominated spaces, especially at night. The presence of men often intimidates women out of fear of harassment or "male mockery of the ability of women engaged in sporting or leisure activities such as running or playing pool" (Valentine, 1989). This theoretical viewpoint holds up in practice; a qualitative study exploring how young African American women perceive and engage with recreation spaces reported that the safety concerns of young adolescent women surrounding outdoor facilities included "disrespectful people, harassment, and fights." The girls in the study claimed that young men dominate recreational sports facilities. These concerns were supported by the researchers, observing young women who did try to play with the young men on the basketball court get ignored. Though, most young women are uninterested in even trying to engage (Ries et al., 2008). Another study found a similar trend in concerns for physical and social safety in determining the likelihood of a teenage girl using a given park for physical activity: Safety reflections like poor lighting and fears of being teased were associated with "a 40% and 35% higher chance of non engagement in PAP, respectively." There was also a notable trend of traditionally male dominated amenities, like skateboard parks, being negatively associated with teenage girls' park use for physical activity, likely related to the intimidation and male-domination that accompanies those amenities (Reis et al., 2009; Yorkshire Sport Foundation, 2022). A case study in Vienna breaks down this dynamic,

explaining that teenage boys and girls often have differing leisure interests which results in then using public space differently and subsequent conflict in that space, and “social conflicts that are usually ‘solved’ on the principle of survival of the fittest,” *the fittest* referring to teenage boys in this case (Stummvoll, 2004).

Whether safety concerns stemming from male domination and harassment affect teenage gender minorities’ park use in the same way that they affect teenage girls has yet to be explored in the literature. However, evidence suggests that concerns about safety, intimidation, and harassment also plague gender minorities in public space, albeit in a slightly different manner. Bornioli et al. (2024) explain that, unlike the sexually objectifying nature of female discomfort in public, gender minorities often experience hostility toward their presentation, such as through aggressive stares. Similarly, Dr. Jody Herman (2013) discusses how the legal or social enforcement of gender segregation in public restrooms negatively affects the extent to which gender minorities can participate in public life.

Interests Unaccounted For. Another common issue is that teenage girls' interests are generally overlooked by designers. The majority of amenities allocated for adolescents tend to be stereotypically male-dominated, such as basketball courts and skateparks. Furthermore, teenage girls do not always use amenities in the ways designers expect. In an experiment where designers were asked to design amenities targeted toward specific age and gender groups, teenage girls made heavy use of the swings designed for them, which typically are not designed for teenage girls, as well as interactive play structures like small green rubber hills with trampolines on top, a stripe painted carousel, and wooden climbing frames which were all designed for young children. However, teenage girls rarely made use of the basketball courts or skateparks compared

to their male counterparts (Lindberg & Schipperijn, 2015). Many girls are also drawn toward playgrounds as preferred park facilities, “yet, many playgrounds are geared toward younger children and have little to offer children in this group (10–13 years old),” (Loukaitou-Sideris & Sideris, 2015; Rivera et al., 2022).

2.2.3 Practical Reasons for Motivating Park Use

Disparities in Physical Activity. Concerns about young women’s underutilization of neighborhood parks often arise from the well-studied range of social and physical health benefits associated with park use. Several studies measuring physical activity across gender and age groups have reported significant disparities between girls and boys, particularly as they approach adulthood.

Using motion detector data, one study monitored physical activity among children ages 10 to 16 and found a heavy correlation between physical activity level with age and sex,

“Before age 13 years, similar levels of physical activity were present in girls and boys ($P = .43$); however, after age 13, boys were significantly more active than were girls (total activity time: $23.5\% \pm 4.7\%$ vs $19.0\% \pm 3.8\%$; $P < .05$).”

They also identified a decline in overall physical activity for both sexes but concluded that this drop-off was more pronounced in teenage girls. Preteen girls spend 35% more time engaged in physical activity than teenage girls (Strauss et al., 2021). This is consistent with other findings indicating that girls ages 8-12 engage in nearly 20% less physical activity than their male peers, with their activity levels steadily declining until age 12 (Telford et al., 2016).

The comparatively lower physical activity levels of teenage girls – to both boys of the same age group and younger girls – have prompted inquiries into the role public parks may play

in encouraging physical activity among young people. An interview-based study in Los Angeles parks found that while participants cited the park as the place where they exercised most frequently, two thirds of those observed engaging in physical activity were men. Additionally, researchers noted “large disparities in park use between boys and girls in both organized and nonorganized activities,” pointing to the need for more facilities designed for women's needs and interests as a way to encourage greater park-based physical activity among women (Cohen et al., 2007). As previously discussed, much of this disparity can be attributed to concerns of harassment, peer pressure, and poor safety (perceived by the girl and/or her parents) (Dwyer et al., 2006; Reis et al., 2009), which can be mitigated by specific design adaptations, which I will discuss in the next section.

Social Belonging and Cognitive Wellbeing. A vast body of literature attributes benefits such as social connectedness, inclusion, and cognitive well-being to park use. In an effort to understand how public green spaces facilitate social inclusion, a study conducted in Zurich found that, regardless of gender, adolescents viewed outdoor spaces like parks and playgrounds as optimal spaces to make friends outside of school. They also identified a positive correlation between age and the use of public green spaces in parks as primarily social spaces, reinforcing the importance of parks in adolescent social development. The researchers also argued that spaces like public parks can act as platforms for “breaking up social segregation,” thereby fostering cultural understanding and exchange: “Open space can be a catalyst for cross-cultural coexistence,” (Seeland et al., 2009).

Researchers at Wageningen University in the Netherlands also expounded upon the role of neighborhood parks in promoting social cohesion, emphasizing their function as a civic space

for people of different cultural and ethnic backgrounds to intermingle (Peters et al., 2010). Beyond socialization, a myriad of evidence speaks to the cognitive benefits of park use, connecting outdoor engagement to improved mental wellbeing, reduced levels of depression and anxiety, and better mood (Zhang et al., 2020; Bustamante et al., 2022). If teenage girls do not visit neighborhood parks, they lose out on the social development opportunities and mental health benefits these spaces offer. This underscores the need for further exploration into strategies that could support teenage girls' use of urban and suburban green spaces.

2.2.4 Planning and Design Strategies for Increasing Usage

How do teenage girls and gender minorities use parks differently? Encouraging neighborhood park use among young women typically requires design adaptations, as teenage girls use parks differently than other demographic groups. Studies examining the nature of adolescent park use often refer to independent socialization among teenagers “even when those parks do not offer young people adequate facilities for recreation.” Researchers consistently note that the social aspect of park use is especially important to young girls, adding that teenage girls are largely attracted to parks with public seating and other areas conducive to passive recreation and socialization (Mahdiar & Dali, 2016; Bornioli et al., 2024).

A United States park use survey reported gender-differentiated dynamics in facility use. In childhood, female children dominated age-specific amenities like playgrounds, but as teenagers, boys began to dominate age-specific spaces like basketball courts and baseball fields, whose girls passively retired to lawns and sidewalks (Evanson et al., 2019). The specific recreational preferences of teenage gender minorities remain largely unexamined in the literature, but it is reasonable to conclude that amenities designed for stereotypically male

2. Literature Review

recreational preferences may not appeal to everyone who deviates from a stereotypically male identity. Understanding how teenage girls and gender minorities use neighborhood parks differently may assist designers in developing targeted amenities that encourage engagement.

Addressing Safety Concerns. As discussed earlier, one of the most significant barriers to teenage girls' park use revolves around safety concerns – both their own and those of their parents. Modifications to park landscapes and management practices may help alleviate these concerns. The literature commonly cited improvements related to visibility and maintenance are commonly cited as key interventions (Valentine, 1989; Mertens et al., 2019). In one survey, women indicated that lighting and equipment maintenance were the most important factors defining their sense of security parks (Polko & Kimic, 2022). Another sample obtained similar results, concluding that planning interventions such as improved lighting, enhancing the layout and cleanliness of bathrooms, removing graffiti (a sign of urban neglect), redesigning green spaces to improve visibility, and replacing solid walls all significantly enhance women's sense of safety in public spaces (Navarrete-Hernandez & Concha, 2021).

A case-study on Vienna's approach to gender sensitive park design highlighted several design practices that appear in the city's planning guidelines. These include lighting "sufficient to allow people to recognise others' faces"; visibility between "interior and exterior zones"; well-maintained facilities that discourage vandalism and "facilitate 'appropriation'"; and, gender-specific playgrounds to reduce spatial conflicts, which tend to favor male-dominated uses of space. The study also emphasized systemic solutions like the intentional recruitment of female planners and architects, while advising against "badly designed, hidden front doors, abrupt

changes from light to dark zones, porches, posts, corners and nooks [which] curtail good visibility” (Valentine, 1989).

Designing for Interest. Speaking to the claims made in the Vienna study, conflict and social tension are often a result of contradictory recreational preferences and the competitive use of space that follows. To encourage park use among young women, it is essential to consider and cater to their preferences. These preferences include interactive recreation facilities like playground and fitness equipment, which are significantly associated with higher levels of park visitation and park-based physical activity among young women (Mertens et al., 2019). Some other interactive play amenities mentioned earlier are incredibly popular among teenage girls. These include trampolines, wooden climbing frames, and swings – girls love swings (Lindberg & Schipperijn, 2015). In terms of methodology, many organizations advocate for a participatory approach to park planning. A case study examining the park equity process in California identified the most effective strategies centered around opportunities for community members to provide input in the park planning and design process and

“co-production of programmes through partnerships with communities and an emphasis on local knowledge and public participation, regular monitoring and evaluation of practices, and the use of new technologies, including social networks,”
(Gibson et al., 2019).

2.3 Reconciling Climate, Public Space Design, and Gender

2.3.1 Climate Considerations for Neighborhood Park Design

Recently, planners have turned to urban and landscape design methods to mitigate the impact of extreme heat on park users. One survey of planners found that the most popular design strategy was the incorporation of urban forests and vegetation, followed by weatherization and

manmade shade (Meerow & Keith, 2022). Another study, which focused on thermal comfort specifically on neighborhood parks in the Phoenix Metropolitan area, found that shade structures like trees and ramadas “*significantly improved outdoor human thermal comfort compared with a full sun exposure,*” however, the degree of relief varied by tree type with some being more effective than artificial shade structures and others being less effective (Colter et al., 2019). Due to the effectiveness of vegetation in reducing heat, parks themselves are frequently used as planning-based heat mitigation strategies (Declet-Barreto et al., 2013; Brown et al., 2015). Beyond cooling, urban parks provide ecosystem services, improving air quality, reduce energy consumption in buildings, and offset carbon emissions (Kim & Coseo, 2018).

2.3.2 The Intersectionality of Climate and Gender

Biological Disparities in Extreme Heat. As extreme heat events become more frequent around the world, there is a growing body of evidence linking heat related hospital admissions and mortality to sex. Studies suggest that women are more likely than men to be admitted to a hospital for respiratory causes in cases of extreme heat (Diaz et al., 2018). While this study found no significant correlation between gender and heat-related hospital admission for circulatory causes, a different study analyzing heat-related mortality reported that “females of all ages had elevated relative risk for deaths from cardiac disease/stroke above threshold temperatures”. Other studies report that, as temperatures exceed threshold, mortality rates for women overall were more than 50% higher than that of men. The disparity was most pronounced among older adults (Folkerts et al., 2022). Climate conditions in general may also affect women’s engagement with the outdoors. Women have reported that factors like temperature,

rain, and seasonal daylight changes act as barriers to outdoor physical activity in public spaces (Hoebeke, 2008; Yorkshire Sport Foundation, 2022).

Extreme Heat, Gender, and Socioeconomic Predispositions. Beyond biological differences in heat vulnerability, intersections between gender, socioeconomic status, and livelihood place women and gender minorities at an increased risk for bearing the burden of climate change compared to men. As it was previously established that heat mortality is feminized, exposure to heat is also feminized, there is also evidence to suggest that in some regions, high surface temperatures may also be feminized. This largely has to do with the fact that urban areas have significantly higher surface temperatures than rural areas due to the urban heat island effect, and the higher representation of women and gender minorities in the service sector more often places them more often in highly-urbanized areas (Goldsmith & Bell, 2021). Further, higher poverty rates among women correspond with an “overexposure of women living in low-value properties that reflects the lack of water bodies, green spaces and trees” in their area (Mashhoodi, 2021). Other scholars extend this to argue that climate change as a whole is feminized, for, women, especially poor women, are the most vulnerable to climate change as their livelihoods are heavily dependent on natural resources like obtaining clean water and firewood necessary for household maintenance. Water contamination and shortages in Africa also disproportionately affect women, who are more likely to live in water-stressed areas. Gendered division of labor and the feminization of poverty leave women in rural areas around the world more vulnerable to indoor air pollution, water-borne illnesses, malaria, and food shortages than men (Denton, 2002). Since women are disproportionately exposed to climate-related ills, any effort to build spaces that are more climate resilient is inherently gender

sensitive. In this case of extreme heat, heat-adaptive neighborhood park design in parks that may be disproportionately used by women, is gender inclusive planning.

Climate Challenges for Gender Minorities. Many of the socioeconomic and health-related challenges that transgender and non-binary people face make them uniquely vulnerable to climate change. According to a study published in the American Journal of Public Health, transgender and non-binary people struggle to access healthcare and are often denied aid during environmental disasters due to both systemic and direct discrimination (Goldsmith & Bell, 2021). Transgender people also experience poverty at more than twice the rate of the general population, which is associated with an overexposure to environmental pollution (Goldsmith & Bell, 2021). Air pollution exposure is especially damaging to gender minorities as air pollution can exacerbate the effects of anxiety and depression in addition to causing respiratory issues for individuals with HIV, all conditions which are far more common in gender minorities than members of the general population (Avara-Meca et al., 2015; Vert et al., 2017). Hence, designing parks with their role in reducing surrounding air pollution in mind is one way in which urban design can consider the needs of gender minorities (Yin et al., 2011).

2.3.3 Gender Inclusion in Climate Discourse

Despite being the most affected by climate change, women and gender minorities are largely excluded from climate change discussions, a debate which “has not sought to address the existing marginalization of women, nor their need to be integrated in environmental policies” (Denton, 2002). Evidently, as climatic extremes continue to influence how we use and design our public recreation spaces, we approach a condition where women and gender minorities are left out of both the climate and public space design conversations. Women and gender minorities

2. Literature Review

suffer disproportionately from the effects of climate change by all metrics, thus, when there is discussion surrounding design-based climate solutions in a given space, it is essential that women and gender minorities are placed at the center of that conversation. Similarly, if we are discussing making spaces more equitable for women and gender minorities, climate adaptation must be a central part of that discussion.

It is fundamental to the integrity of the gender sensitive park design discourse to place it in conversation with climate sensitive park design. My research brings climate into the gender conversation and gender into the climate conversation. By exploring the park use of teenage girls and gender minorities – in part, through the lens of extreme heat –, and using a participant-centered methodology to develop neighborhood park design strategies that are specifically tailored to heat resilience, I develop a working framework for the relationship between gender, climate, and park design.

Methodology

As discussed in Section 1 Background and Section 2 Literature Review, there is a growing conversation pertaining to the unique ways in which women, girls, and gender minorities engage with public space (Bornioli et al., 2024). There is also a myriad of research describing how climate change shapes the ways that people perceive and engage with public space. Much of the research in both of these areas is solutions-driven. Even though a magnitude of literature unpacks how climate change disparately affects women and gender minorities, the conversation about the effects of climate change on young women and gender minorities' public space use, particularly in hot desert cities in a Global North context, is still in its infant state. Hence, this research initiative focuses on adolescent girls and gender minorities in the Phoenix, Arizona metro area – the most populous hot desert city in the United States and the Global North.

Furthermore, it is practically consensus that the best way to design public spaces that work better for teenage girls and gender minorities is to simply ask them what they want (Walker & Clark, 2023). What do they want to change, keep, or add? To reflect this best practice, the data collection process was designed to provide participants with as many opportunities as possible to share both prompted and unprompted reflections about their relationship with their local neighborhood parks and their opinions regarding a specific neighborhood park in Tempe, Arizona (Phoenix metro area).

3.1 Research Questions

This research initiative seeks to answer the following questions:

1. How do adolescent girls and gender minorities use, perceive, and feel about their neighborhood parks in the Phoenix metro area?
 - a. How do these feelings and interactions evolve from childhood to adolescence?
 - b. How does Phoenix's hot desert climate affect the neighborhood park use of teenage girls and gender minorities?
2. How do adolescent girls and gender minorities in the Phoenix metro area perceive the quality of existing neighborhood parks?
3. What steps can urban stakeholders take to plan and design neighborhood parks that serve teenage girls and gender minorities while also striving for climate resilience?

3.2 Process Overview

To answer these questions in a way that places adolescent girls and gender minorities at the center of this research, I employ a modified participatory action methodology. This methodological framework is based on the *Her City Toolbox*, which was designed by UN-Habitat as a guide for participatory public space planning and design. The *Her City Toolbox* includes 9 Flexible Building Blocks for development, but this research narrows in on Blocks 2-6: City wide assessment (3.4 Survey Methodology), Site specific assessment (3.5.3 Observations Template), Analyzing challenges (3.5.4 Reflection; 3.5.5 Solutions Discussion), Designing ideas (3.5.6 Solutions Design Activity), and Recommendations for action (6.2 Recommendations) (UN-Habitat, 2021).

3. Methodology

I partnered with The Girl Scouts – Arizona Cactus-Pine Council to recruit 55 Girl Scouts between the ages of 11 and 18 years. Every Girl Scout who registered for this event was a resident of Maricopa County, which encompasses the Phoenix metro area. Each Girl Scout participated in two phases of data collection, in the following order: 1) an individual survey and 2) a workshop. The goal of the individual survey was to assess the Girl Scouts’ personal past and present relationships with neighborhood parks throughout the Phoenix metro area (See 3.4 Survey). After completing the survey, they attended a workshop at Hudson Park (Parque de Soza) on November 12th, 2022. During the workshop, the Girl Scouts worked in groups and as individuals to conduct a site-specific assessment of Hudson Park. This included various educational interventions (See 3.5.2 Workshop Introduction, 3.5.4 Reflection, and 3.5.5 Solutions Discussion), a small-group activity where participants made different observations about the park (See 3.5.3 Observations Template), and an individual activity where the Girl Scouts redesigned Hudson Park in a way that better suited their needs and interests (See 3.5.6 Solutions Design Activity).

Once both data collection phases were complete, the data from the survey and workshop were analyzed separately. Descriptive statistics and thematic coding results from both phases of data collection are reported in Section 4 Survey Results and Discussion and Section 5 Workshop Results and Discussion. Then, the findings from both phases were reviewed holistically to design recommendations for hot desert city neighborhood park design best practices in a Global North context. Figure 1 provides a visual representation of this process that is specific to this research initiative. A generalized version of this process is available in Appendix A.

3. Methodology

My research deviates from traditional definitions of Participatory Action Research (PAR) methodology, as it does not involve the adolescent girls and gender minorities in the framing of the research itself (Carleton College, n.d.). However, the participants do lead the process in the sense that they collect the Hudson Park quality data (See 3.5.3 Observations Template); they identify areas of concern (See 3.5.4 Reflection, 3.5.5 Solutions Discussion); they design solutions to address those concerns (See 3.5.6 Solutions Design); and the feedback, input, and solution designs that the adolescent girls and gender minorities provided throughout all aspects of the data collection portion of the project are the subject of the recommendations I present (See 6.2 Recommendations) in their entirety.

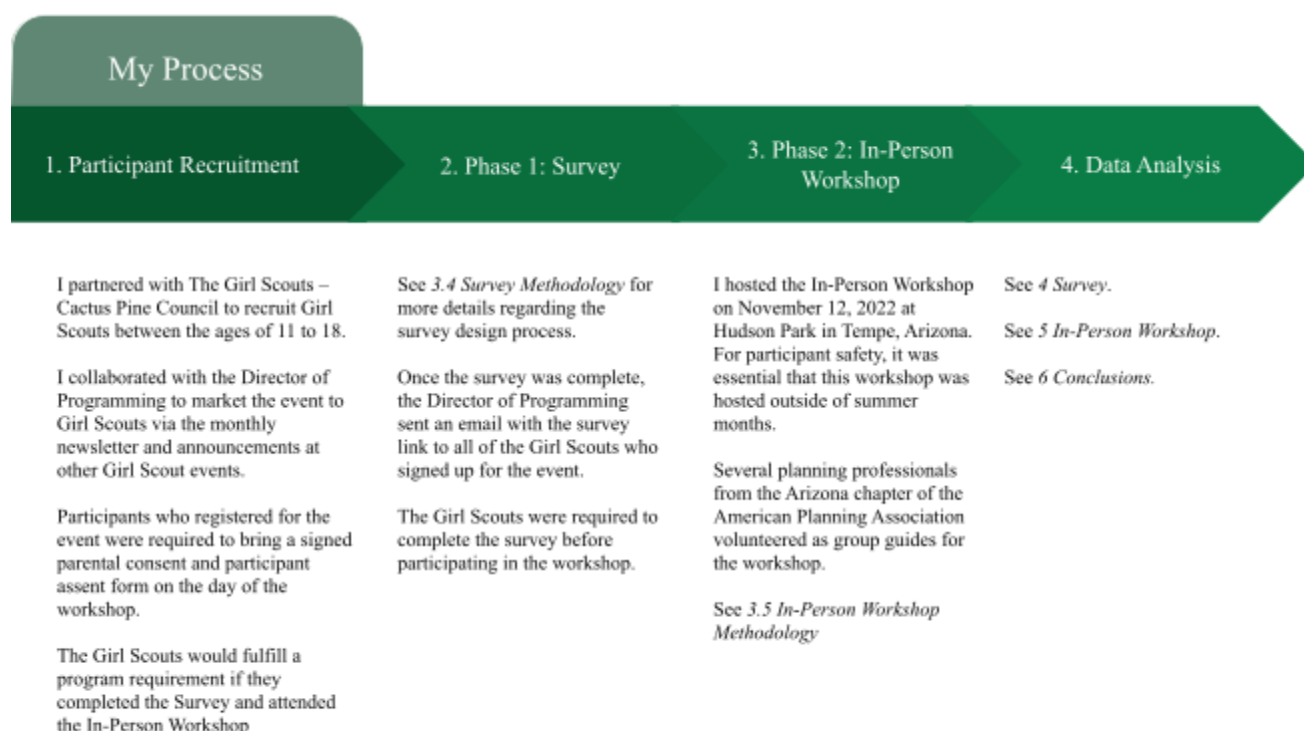


Figure 1. A visual depiction of the research methodology.

3.3 Sample Description

The sample is made up of teenagers who identify either as girls or as gender minorities. For ease of communication, I will be referring to those participants as TGM for the remainder of the thesis. From this point forward, unless otherwise specified, “park(s)”, “public park(s)”, and “local park(s)” will all refer to “neighborhood park(s)”. For this initiative, a *teenager/teen* referred to the participant at their current age (i.e. anyone from 11 to 18 years of age).

3.4 Survey Methodology

Upon registering for the workshop, the TGM were sent a link to a survey (See Appendix B), which they were instructed to complete before attending the workshop. The survey included questions related to the participants’ demographics (age, gender identity, LGBTQIA+ identity, county of residence), their perceived childhood and present neighborhood park use, their perceptions of the influence of weather and climate on their neighborhood park use, and their general feelings about their neighborhood parks.

I received 54 survey responses. After eliminating observations with less than 50% progress, 42 observations remained for analysis. As depicted in Table 1, the vast majority of respondents were between the ages of 11 and 16, with a fairly even distribution within this range: 11-12 (33.33%), 13-14 (28.57%), 15-16 (30.95%). Fewer than 10% of the TGM were between the ages of 17 and 18 (7.14%). A little over a third of the 41 respondents who reported their LGBTQ+ status (36.59%) indicated that they identified as a member of the LGBTQ+ community. The rest of the respondents either did not identify (46.34%) or were not sure of their identity (17.07%). In terms of gender identity, most respondents identified as female (80.49%),

3. Methodology

six identified as non-binary or third gender (14.63%), and two identified as male (<5%). Since Girl Scouts restricts its membership to cisgender girls, gender-expansive, non-binary, intersex, or transgender people, these male-identifying observations remained in the sample as they were deemed to be either a mistake or indicative of a trans-male identity. All respondents resided in Maricopa County.

Variable	Category	Count	Percentage (%)
Age	11-12	14	33.33
	13-14	12	28.57
	15-16	13	30.95
	17-18	3	7.14
LGBTQ+ Status	Identifies	15	36.59
	Does not Identify	19	46.34
	Not sure	7	17.01
Gender	Female	33	80.49
	Non-binary or Third Gender	6	14.63
	Male	2	4.87
County of Residence	Maricopa	42	100
	Other	0	0
Sample Size		42	

Table 1: Demographic breakdown of survey respondents

3.5 Workshop Methodology

After the survey phase of data collection, 55 TGM attended the workshop on November 12th, 2022. In addition to the TGM, eight local volunteer planning professionals attended the event.

3.5.1 Pre-Workshop

As they arrived, each TGM provided the Girl Scout–Cactus Pine Council’s Director of Programming with a signed parental consent form. If they did not complete the survey before the workshop began, they were provided with a paper copy to complete the same day. Each attendee was provided with a name tag that specified their group. Groups were assigned based on *troop number*. If someone was attending the workshop without a *troop*, they were placed in a group with members of a similar age. There were eight groups in total, and I assigned a volunteer planning professional to each group. When the workshop began, I gathered the attendees into the park amphitheater to go over the agenda and begin programming for the day.

3.5.2 Workshop Introduction

First, I introduced myself to the TGM. Then, I told the TGM a story about a participatory action initiative in Malmö, Sweden that consulted teenage girls in the development of a public recreation space. The TGM were then asked to share how frequently they visit their neighborhood park. After fielding a few responses, I shared the following statistic:

“Boys over the age of 8 use parks four times as much as girls. And that more than 40% of girls ages 11 to 21 said they feel unsafe outdoors.”

I contextualized the statistic by sharing the fact that, historically, the majority of urban planners have been male and asked the rhetorical question, “How can you design cities for everyone to live in, if the people living in that city are not being fully represented by the people planning them?”

Then, I defined and described the concept of “Urban Planning” as “the process of developing and designing urban areas to meet the needs of a community.” I explained that the

urban planning method used in the Malmö story was “Participatory Action Research” or PAR. I defined PAR as

“a framework for conducting research and generating knowledge centered on the belief that those who are most impacted by research should be the ones taking the lead in framing the questions, the design, methods, and the modes of analysis of such research projects.”

After defining PAR, I communicated to the TGM that, by participating in this workshop, they are also taking part in a form of PAR (See 3.2 Process Overview). I provided the participants with the following instruction:

“Today, you are going to be the researcher. Oftentimes, when a researcher tries to collect information, how they feel about that information doesn’t matter. In this case, what you feel about the information you collect is what is most important. You are the girls who we need to design for. There are no wrong answers. You decide what the right answers are and it’s our job as the planners and researchers to get that right.”

I concluded the introduction by sharing the rest of the agenda for the workshop.

3.5.3 Observations Template

After the introduction, the TGM split into their assigned groups. Each group was paired with one of the 8 volunteers and provided an observations template complete as a group. The TGM were instructed to take the lead on the activity and to refer to their assigned volunteer for guidance as needed.

The Observations Template (Appendix C) was designed to assess the quality of Hudson Park (Parque de Soza). It combines themes and questions from an extension of the Public Space Index for mid-sized parks (Evans et al., 2018) and “Measuring Park Quality for Youth 13-19” (Kocmaruk, 2021). The observations template prompted the TGM to make both qualitative and quantitative observations about the park. It also asked the TGM to reflect on how they feel about certain aspects of the park as well as the types of activities they could imagine themselves

conducting in the park. With respect to the Public Space Index, a public space's quality can be assessed by determining the extent to which it adheres to five dimensions: *inclusiveness*, *meaningfulness*, *safety*, *comfort* and *pleasurability* (Mehta, 2014). Thus, to use the results of the observations template in effect of a site-specific assessment, each question and prompt was categorized as evaluating one of the following elements of the park: *Design and Aesthetic*, *Comfort (Safety)*, *Comfort (Maintenance and Environment)*, and *Activities*.

3.5.4 Reflection

After completing the observations template, each group was given pieces of paper with feeling statements like “I would avoid this part of the park if other people were here in general” and “I would hang out with friends in this part of the park.” They were also given smaller pieces of paper with the different areas of the park (e.g. “The Ramada”, “The Volleyball Courts”, etc...) written on them. The TGM were instructed to reflect on the observations activity as a group, and with their assigned expert volunteer, to determine which parts of the park they would associate with each prompt.

The TGM and volunteers were dismissed for a 15 minute break before being called back to the amphitheater to discuss the observations template and prompt matching reflections. In an open forum, I asked the attendees the following questions:

1. What did we notice about the park from the observation activity?
2. Were there any questions from the prompt matching activity that you thought were odd or confusing? Did any questions provoke you to think about something you may not have previously considered?

3. What were the characteristics of places you felt positively about? What about those you felt negatively about?
4. Which park did you match to each prompt?

3.5.5 Solutions Discussion

Activity reflection moved into a discussion about potential solutions that could be implemented to address problematic aspects of the park or to make it more enjoyable for them. First, I asked the attendees to share ideas for solutions and improvements. Then, I told the attendees a story about an art installation in Boston, Massachusetts called *Swingtime* that was particularly popular with teenage girls (Lange, 2021). Lastly, we discussed other potential solutions to help TGM have more fun and feel more comfortable in their neighborhood parks. To prompt conversation, I proposed some of the solution recommendations presented in the Make Space for Girls (n.d.b) report “Better Design Suggestion for Parks” and the Yorkshire Sport Foundation (2022) “Make Space for Us” Insight Report. Examples of the solutions I shared included shaded places to hang out, expanding green spaces, installing age-inclusive swings, enhancing lighting, subdividing sports areas, improving bathroom maintenance, and installing seating areas arranged in groups rather than lines.

3.5.6 Solutions Design Activity

Attendees had the opportunity to expand on the solutions discussion by completing a solutions design activity while the expert volunteers remained available for assistance. I presented them with concept visuals from previous gender sensitive public park design initiatives (Appendix D) as inspiration and encouraged them to reflect on the aspects of the concepts that did and did not appeal to them. I provided each TGM with their own animated map of the park

3. Methodology

(Appendix E). Then, I instructed them to cross out features that they would remove from the park, note features that they would change, and draw features that they would add. Once participants completed the activity, I took a picture of their finished map, and then they were allowed to take it home with them.

In the remaining portion of this thesis, I will report and discuss the results from the survey (See 4. Survey Results and Analysis) and the workshop (See 5. Workshop Results and Analysis). Then, I will consolidate the research conducted in the literature review with my findings to draw conclusions and deliver recommendations regarding effective approaches to gender inclusive and climate sensitive park design.

Survey Results and Discussion

Analyzing the TGM's past, present, and desired relationship with neighborhood parks is a necessary prerequisite to designing parks that work for them. In this chapter, I will compare the TGM's past and present park use, parse the effect of extreme heat on park visitation, call out the qualities of parks in the Phoenix metro area that currently appeal to the TGM, and characterize the changes that the TGM would like to see in their own neighborhood parks. The results discussed in this chapter will establish the reasons why and extent to which Phoenix metro area parks are failing to meet the needs and interest of teenage girls and gender minorities.

4.1 Evaluating Neighborhood Park Use

A significant portion of the survey questions asked the TGM to recall different aspects of their park use as a child and compare it to their present park. They selected the frequency at which they recall visiting and the activities they conduct(ed) in the park as both a child and a teenager. Several of the TGM whose park use changed over time also shared reflections as to why their relationship with their local park has changed.

4.1.1 Visitation Frequency

All 42 respondents reported their current and childhood perceived neighborhood park use frequency (Table 2). As teenagers, only two respondents (4.76%) report visiting neighborhood park(s) multiple times per week. One-third (N = 14) report visiting neighborhood park(s) a few times per month. The plurality (42.86%) report visiting neighborhood park(s) only a few times

4. Survey Results and Discussion

per year. Nearly a fifth (19.05%, N = 8) report visiting neighborhood park(s) less than once per year (Table 1). Of the two who reported visiting the park “Frequently (Multiple times per week)”, both were 13 years old or younger. Exactly one-third of participants aged 14 or older reported visiting the park “Never or Almost Never (Less than once per year)” as an adolescent, while only two participants ages 13 or younger (8.33%) reported the same. This indicates that adolescent girls and non-binary people may visit the park less and less as they progress through adolescence; however, the sample size is too small to draw that conclusion definitively.

Childhood vs. Adolescence. Overall, participant survey responses demonstrate a drastic decrease in the frequency of park visitation from childhood to adolescence. Nearly half (47.62%, N = 20) report visiting neighborhood park(s) multiple times per week when they were children, and another 45% (N = 19) report visiting a few times per month. Only three respondents (less than 8%) report having visited neighborhood park(s) a few times per year or less (Table 1).

	Child		Teenager/Adolescent	
Perceived Visitation Frequency	Count	Percent	Count	Percent
Frequently (Multiple times per week)	20	47.62	2	4.76
Occasionally (A few times per month)	19	45.24	14	33.33
Rarely (A few times per year)	1	2.38	18	42.86
Never or Almost Never (Less than once per year)	2	4.76	8	19.05
Total	42	100.00	42	100.00

Table 2. Childhood vs. Adolescent comparison of neighborhood park visitation frequency distribution

Comparing reported childhood neighborhood park visitation frequencies with current reported frequencies, it is evident that neighborhood park visitation decreased significantly among the TGM as they entered adolescence. For this analysis, frequency categories were coded

4. Survey Results and Discussion

ordinally (1=“Never or Almost Never”, 2=“Rarely”, 3=“Occasionally”, 4=“Frequently”), such that an n-unit decrease in neighborhood park visitation frequency indicates that a participant reported being in a neighborhood park visitation category as a child coded as n-units more than the one in which they reported being as a teen. No respondents reported using neighborhood parks more as adolescents/teens than as children. They all either stayed in the same visitation frequency category (19.05%, N=8) or reported decreasing their use frequency. Only six of the 42 respondents used the park with a meaningful frequency (not “Never or Almost Never”) as a child and continued to use it at that same category of frequency. More than half of the respondents (57.14%, N=24) reported that their neighborhood park visitation frequency decreased by one category (e.g. “Frequently” to “Occasionally”, “Occasionally” to “Rarely”, etc...) from childhood to adolescence. Nearly one-fourth (23.81%, N=10) indicated a neighborhood park visitation frequency decrease of two or more categories. I conducted a Wilcoxon Signed-Rank Test to conclude, with 99.99% confidence ($P\text{-Value}=0.0000$), that the Girl Scouts’ self-perceived neighborhood park visitation frequency decreased from childhood to adolescence (Appendix F.1). Overall, the data exhibits a marked drop-off in girls’ and gender minorities’ perceived neighborhood park visitation as they transition from childhood to adolescence.

Reasons for Decreased and Infrequent Neighborhood Park Visitation. Of the Girl Scouts who reported visiting neighborhood parks less frequently as a teenager, 34 provided additional insight as to why that was the case (Figure 1). Respondents also shared information about their neighborhood park perceptions and feelings that may be used to draw implicit conclusions about discontinued park use.

4. Survey Results and Discussion

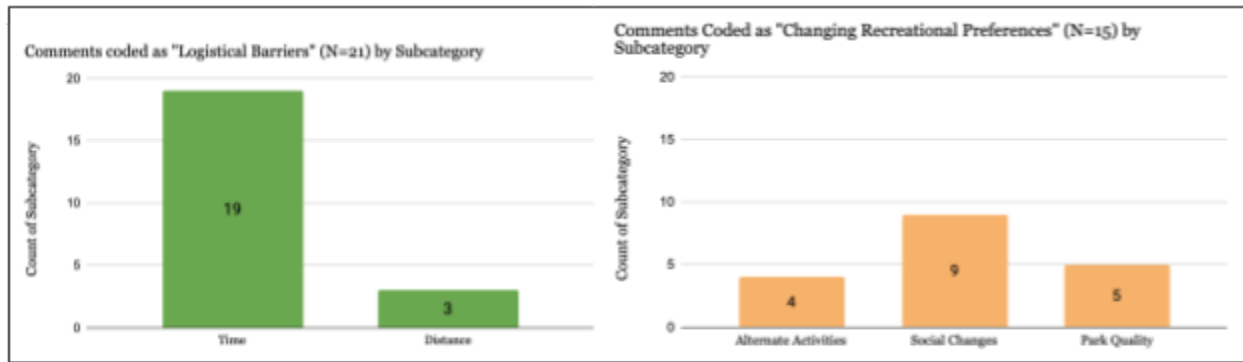


Figure 2. Side-by-side bar charts comparing counts of most commonly cited reasons for reporting less frequent or discontinued park use as an adolescent compared to as a child, broken down by subcategory.

Logistical barriers (temporal and spatial). Logistical barriers, ranging from schedule constraints to an inability to access transportation, were by far the most common factors inhibiting park visitation frequency. Of the 34 Girl Scouts who explained why they visit neighborhood parks less as a teenager, over two-thirds (N=21) cited logistical barriers as a key reason why they stopped visiting the park (Figure 1, left). In particular, 19 of the respondents indicated that the increasing demands of school and extracurricular activities made it harder to find time to visit neighborhood parks. Three of the respondents also specified that the park was too far given their available means of feasible transportation. Simply, “There’s nobody to take me.” Neighborhood park location, specifically proximity to frequented spaces like home and school, appears to be important to the TGM. Of the 24 TGM who reported having a favorite neighborhood park, 18 (75%) of them liked that their favorite neighborhood park was close to their home or school. Not only were time and distance-related constraints major barriers for TGM park use, but the TGM were favorable toward neighborhood parks that were conveniently located.

4. Survey Results and Discussion

Changing recreational preferences. Many of the respondents also specified that changes in their own recreational preferences or their friends' preferences meant that visiting the park just did not make sense anymore (Figure 2, right). While some Girl Scouts simply preferred other activities like using technology, a few took issue with what the park itself had to offer, or *did not offer*: "I don't find it fun anymore." As teenagers, they felt that there was not much for them to do at the park. Even if they wanted to go, their friends stopped going, so who would they go with? The sentiment of not having much to do at the neighborhood park was shared by other survey takers, with nearly 40% percent of the respondents (N=15) feeling like there is not "plenty for [them] to do" at their neighborhood park (Appendix G). Though, there did appear to be an age divide. A Wilcoxon Rank-Sum test indicated that respondents 14 years old or younger agreed more that there was "plenty for [them] to do" than those older than 14 years. Though, this was only significant at the 92.34% confidence level (Appendix F.2). Still, I do not recommend completely ruling out the possibility that older teens may feel differently about the extent to which their neighborhood parks cater to them than younger teens and pre-teens. Also along age lines, most respondents felt like there is not much for teenagers to do at their neighborhood park (60%, N=23) and that it is "more for little kids than it is for [them]" (80%, N=30, Appendix G.1). As the TGM age, they are less and less interested in the recreational activities that neighborhood parks have to offer them, resulting in reduced park visitation.

A space not being designed for one's age, or demographic group in general, not only makes visiting the space less enjoyable, recreationally, but it also may foster a sense of social discomfort by making one feel like they do not belong in the space (Fortune et al., 2021; Powers et al., 2024). Overall, the Girl Scouts did not feel strongly about their sense of belonging in their

4. Survey Results and Discussion

local neighborhood park (Appendix G.1). However, a Wilcoxon Rank-Sum test revealed that those who were older than 14 years tended to disagree that “At [their] neighborhood park, [they] belong” significantly more ($\text{Prob} > |z| = 0.0137$) than girls who were 14 years old or younger (Appendix F.3). This supports the notion that older teenagers may feel more excluded from and less catered to by their neighborhood parks than younger teenagers and pre-teens.

While a neighborhood park planner or designer may be limited in their ability to mitigate temporal and spatial limitations to adolescent neighborhood park use, there is certainly room to design neighborhood parks that cater to the recreational preferences of adolescent girls and non-binary people in a way that fosters a sense of belonging among teenagers of all ages. Though, it is critical to understand these recreational preferences to do so.

4.1.2 The Nature of Adolescent Park Use

Beyond visitation frequency, I also analyzed the nature of the Girl Scouts’ neighborhood park use. In this chapter, I determine the features that the TGM use, the activities that they carry-out in neighborhood parks, and how those activities have changed over time.

Features Available and Features Used. Understanding the frequency at which the TGM use standard park features, given they have access to those features in their local neighborhood park, can help develop an understanding of their actual feature preferences. The vast majority of respondents reported that their neighborhood park had a playground (Table 3). Of those who reported having access to a playground in their neighborhood park, just over half of them ($N=16$) report using the playground “Often” or “Always”. Swings were also popular; nearly two-thirds of those who reported having access to swings also reported that they use the swings “Often” or “Always” (Table 3). Most girls also had access to picnic tables ($N=31$, 91.18%), the majority of

4. Survey Results and Discussion

whom used them “Often” or “Always” (Table 3). In contrast, those who had access to a tennis, volleyball, or basketball court did not use them very often at all with a majority of the TGM using them “Never” or “Rarely” (Table 3). Overall, features that facilitated passive recreation, like swings and picnic tables, were used more frequently than active recreation features, like sport courts (Table 3).

	Reported Present in Respondent's Neighborhood Park		Percent Use Frequency				
Park Feature	Count	Percent	Never	Rarely	Often	Always	Does Not Apply
Active Recreation Features							
Playground	31	91.18	6	8	10	6	1
Swings	24	70.59	2	7	8	7	0
Tennis Court	7	20.59	4	1	1	1	0
Volleyball Court	10	29.41	2	4	2	2	0
Basketball Court	14	41.18	7	5	2	0	0
Skate Park	1	2.94	0	0	0	1	0
Passive Recreation Features							
Field	27	79.41	4	9	10	3	1
Garden	5	14.71	1	1	1	1	1
Pool/Water Feature	8	23.53	2	3	2	0	1
Picnic Tables	31	91.18	3	7	13	8	0
Total	34						

Table 3. Reported presence of neighborhood park features vs. use-frequency of those features.

4. Survey Results and Discussion

Activities – Adolescence vs. Childhood. Comparing the activities that the TGM reported carrying out in their local parks as children and teenagers can provide insight into how the TGM's relationship with their parks – and recreation in general – have changed over time. Of the TGM who reported whether or not they engaged in certain activities when they visited the park (N=34), 10 reported exercising (29.41%), 11 reported using sport facilities (32.35%), and 15 reported using the playground when they visited their neighborhood park (44.12%). With respect to more social activities, they reported spending time with friends (N=19, 55.88%), engaging in family activities (N=13, 38.24%), and attending community events (N=11, 32.35%). Notably, more than half of the TGM (N=22 out of 39) reported mostly spending time with people of the same gender at the park.

The childhood neighborhood park activities participants could select in the survey were different from the teenage neighborhood park activities. To effectively compare the nature of childhood and adolescent neighborhood park use, I coded the neighborhood park activities that participants engaged in as either *elective* (e.g. playing with friends, playing games like tag, and using recreation features like the pool or playground) or *obligatory* (e.g. attending community or family events, attending classes, playing sports/exercising, or taking care of pets). A Wilcoxon Signed-Rank Test shows that more respondents reported engaging in *elective* activities as children than as teens, significant beyond the 99% confidence level (Appendix F.4). There was not a statistically significant distinction between childhood and adolescence when it came to engaging in *obligatory* neighborhood park activities. The respondents in our sample recall visiting parks as children for both elective recreation and obligation. Now, as teenagers, when they visit the park, the nature of their visit is significantly less likely to be for general enjoyment

4. Survey Results and Discussion

than it is to achieve a specific purpose. While attending a community event or playing an organized sport may still be fun, visiting a neighborhood park because an event happens to be hosted there is distinct from visiting a neighborhood park to simply enjoy the space and what it has to offer.

The playground is one notable exception, however. Many respondents who reported playing on the playground as children (N=40) claim to use the playground in adolescence (Table 4). A chi-squared test of independence did not indicate ($Pr = 0.837$) that respondents in one adolescent age category were any more likely than respondents in any other adolescent age category to report using the playground in the present. In this sample, survey takers of all ages enjoy playing on the playground (Appendix F.5).

	Reports Playground Use as Teenager		
Reports Playground Use as Child	Yes	No	Total
Yes	15	17	32
No	0	2	2
Total	15	19	34

Table 4. Count of TGM who report using the playground as a child vs. a teenager

4.1.3 Summer Heat and Neighborhood Park Use

The Phoenix-metro area is considered the hottest major city in the United States (National Aeronautics and Space Administration, 2022). In 2022, Phoenix saw more than 187 days with highs above 90 degrees Fahrenheit (Extreme Weather Watch, 2025). With summer temperatures frequently climbing above 110 degrees Fahrenheit, outdoor activity is often uncomfortable or even dangerous. The survey data revealed a seasonal component of Phoenix

4. Survey Results and Discussion

metro area neighborhood park use. In particular, extreme heat is a significant barrier to park use within the sample. This may also exacerbate the effect of logistical barriers on neighborhood park visitation frequency.

Seasonality and Extreme Heat. Many cities in the Global North experience cold winters, making winter the season when park visitation is lowest (Roemmich & Johnson, 2014). Though, mild winters, sweltering summers, and the findings from this survey suggest that this is not the case in the Phoenix metro area. The TGM were asked to rank the seasons in order of when they would be likely to visit the park. A total of 37 completed the ranking, where 70.27% (N=26) of respondents ranked summer last, saying they would be least likely to visit the park then. Another 16.22% (N=6) ranked it third, while a mere 13.51% (N=5) ranked it first. In contrast, 83.78% (N=31) of respondents ranked the fall first or second, and 72.99% (N=27) of respondents ranked the spring first or second. Lastly, 27% of respondents ranked the winter last, 43.24% ranked it third, 18.92% ranked it second, and 10.81% ranked it first. Rather than mimicking the typical trend of low park visitation in the winter, the Phoenix-based TGM report the opposite behavior, claiming that they are most likely to visit the park in the fall and springtime and least likely to visit the park in the summertime.

Extreme heat is likely at the core of TGM reporting the least amount of park use during the summer. Considering the extreme summer temperatures of the Phoenix metro area, infrequent neighborhood park visitation during the summer is consistent with other literature denoting 91.4 degrees Fahrenheit as a critical upper threshold before heat-related declines in visitation ensue (Hewer et al., 2016; Extreme Weather Watch, 2025). When asked how high outdoor temperatures affect their decision to visit the park, the vast majority of the respondents

4. Survey Results and Discussion

(89.74%, N=35 out of 39) said that they are less likely to use the park on hot days, while a mere 7.69% (N=3 out of 39) claim that the temperature does not affect their decision to visit the park, and only one respondent indicated being more likely to visit the park on hot days. Moreover, 74.36% (N=27) of the same Girl Scouts said that they “Somewhat Agree” or “Strongly Agree” with the notion that they “would go to the park more in the summer if [they] lived somewhere with cooler summer temperatures.” Unmitigated heat is clearly a major obstacle to summertime neighborhood park use, especially in hot desert cities like Phoenix. This issue only stands to become more pronounced as Arizona summers get hotter and longer each year (Rainson, 2024).

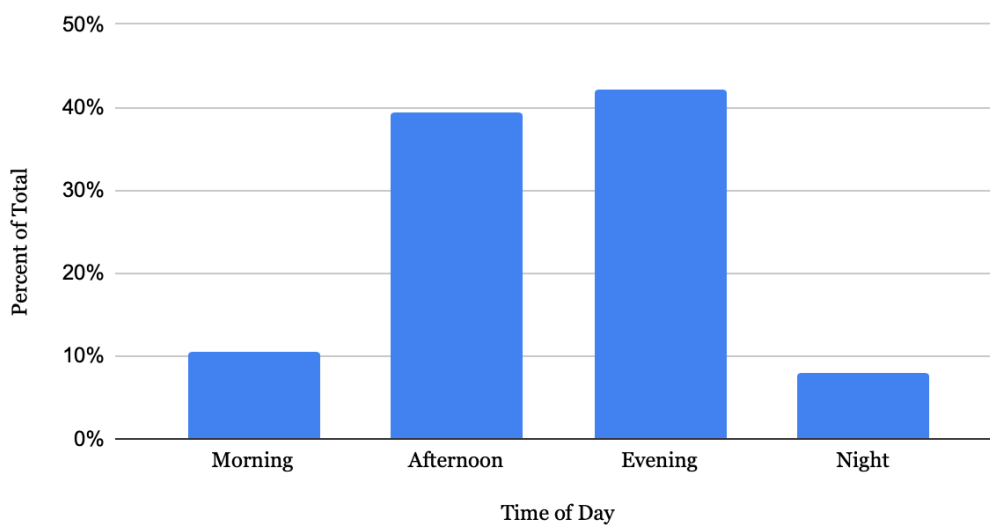
Exacerbating the Effect of Logistical Barriers. With nearly half of the sample explicitly citing time constraints related to school and extracurriculars as a key reason why their neighborhood park visitation has decreased in adolescence, it is natural to reason that they may visit their neighborhood parks more during summer vacation. Yet, neighborhood park visitation is the least frequent in the summertime, even among the respondents expressing woes about busy school years.

To avoid the heat, teenagers could visit the park on summer nights when high temperatures are less pronounced. Still, 31 Girl Scouts indicated that, if they were to visit their neighborhood park during the summer, they would most likely go in the afternoon or evening – the hottest parts of the day (Figure 3). This may be a result of the general discomfort adolescent girls and non-binary people experience while being in the park at night, with poor visibility. In fact, when asked whether they felt comfortable in their neighborhood park at night, more than 40% of the 39 who responded said that they “Somewhat Disagree” or “Strongly Disagree” (Appendix G.2). When asked if they would visit the park more in the summer if they “felt

4. Survey Results and Discussion

comfortable going at night” almost half (48.72%, N=19) of the respondents selected “Somewhat Agree” or “Strongly Agree”. This is consistent with existing research, which claims that women’s urban mobility is relatively restricted at night due to a perceived lack of safety (Condon et al., 2007). A large portion of the TGM feel uncomfortable in the park at night. For many of them, this discomfort is inhibiting their ability to access the park during the summer months.

Figure 3: Time of Day Respondents Report Being Most Likely to Visit Their Neighborhood Park in the Summer



The combination of climate-related discomfort during summer days and safety-related discomfort during summer nights makes going to one’s neighborhood park a non-option for many of the TGM at the time of year when they have the most time for outdoor recreation. For busy teens, the poor timing of extreme summer heat with school-year obligations make parks practically inaccessible year-round. Additionally, this year-round inaccessibility may be somewhat feminized. Although this survey did not sample adolescent boys, based on past research related to gender disparities regarding perception of nighttime safety in urban parks, it is reasonable to assume that teenage boys might feel safer visiting the park at night than their

4. Survey Results and Discussion

teenage girl and female-presenting non-binary counterparts (Polko & Kimic, 2022). Other research also suggests that poor weather is a larger deterrent of park physical activity for girls than boys (Yorkshire Sport Foundation, 2022). Therefore, the effective year-round restriction of neighborhood park access may be less prominent for adolescent boys who might be more likely to visit neighborhood parks on summer nights or in extreme weather, but more research is necessary to draw definitive conclusions.

4.2 Motivating Neighborhood Park Visitation

The survey results discussed thus far have been largely consistent with the existing theory on non-male adolescent park use. Teenage girls are visiting neighborhood parks infrequently, and far less often than they did as children (See 4.1.1 Visitation Frequency; Walker & Clark, 2023). When they do visit the park, they go more out of obligation or to complete a specific task than for general recreation and enjoyment (See 4.1.2 The Nature of Adolescent Park Use). While much of this phenomenon can be attributed to busy schedules, many adolescent girls and non-binary people are simply underwhelmed by what their neighborhood parks have to offer them. There's not much to do there, and, in their minds, their local park is for kids, not teens (See 4.1.1 Visitation Frequency; Walker & Clark, 2023). Weather is also a key factor affecting TGM park visitation (Yorkshire Sport Foundation, 2022) with extreme heat imposing a further logistical burden on the TGM (See 4.1.3 Summer Heat and Neighborhood Park Use). Naturally, increasing neighborhood park visitation among the TGM entails designing parks that are not only comfortable and entertaining, but sufficiently comfortable and entertaining to persuade them to choose visiting the park over alternative leisure activities in their limited free-time.

4. Survey Results and Discussion

The discussion on the park features used and activities conducted by the Girl Scouts in the sample provides some insight into how teenage girls and non-binary people currently use public parks (See 4.1.2 The Nature of Adolescent Park Use). However, simply tweaking neighborhood park designs to accommodate these activities and include more of these features is not sufficient to spur more frequent park visitation, as it describes how the participants use neighborhood parks in their current state, not necessarily how they would use neighborhood parks in their ideal state. Thus, increasing park visitation frequency among adolescent girls and gender minorities requires an understanding how the TGM would use parks in their ideal state and, more importantly, what the ideal state of a park is to them. To achieve this, this section describes 1) the characteristics of parks that the TGM generally favor, 2) the aspects of parks that the TGM take issue with, and 3) the changes the TGM would make to their local park if given the opportunity to do so.

4.2.1 What's Working?

To assess the characteristics of parks that the TGM generally favor, the survey asked them to share whether they had a favorite neighborhood park, the name of the neighborhood park, and the reasons why they liked that park.

More than half of the TGM indicated that they did have a favorite neighborhood park (57.14%, N=24). Of those who responded “Yes”, several shared the name of their favorite park. Figure 4 depicts the distribution of the “favorite parks” throughout the Phoenix metro area, illustrating heavy representation in Northwest, North, Northeast, and East regions of the Phoenix metro area. There was no representation in the West or Central regions of the Phoenix metro area.

4. Survey Results and Discussion

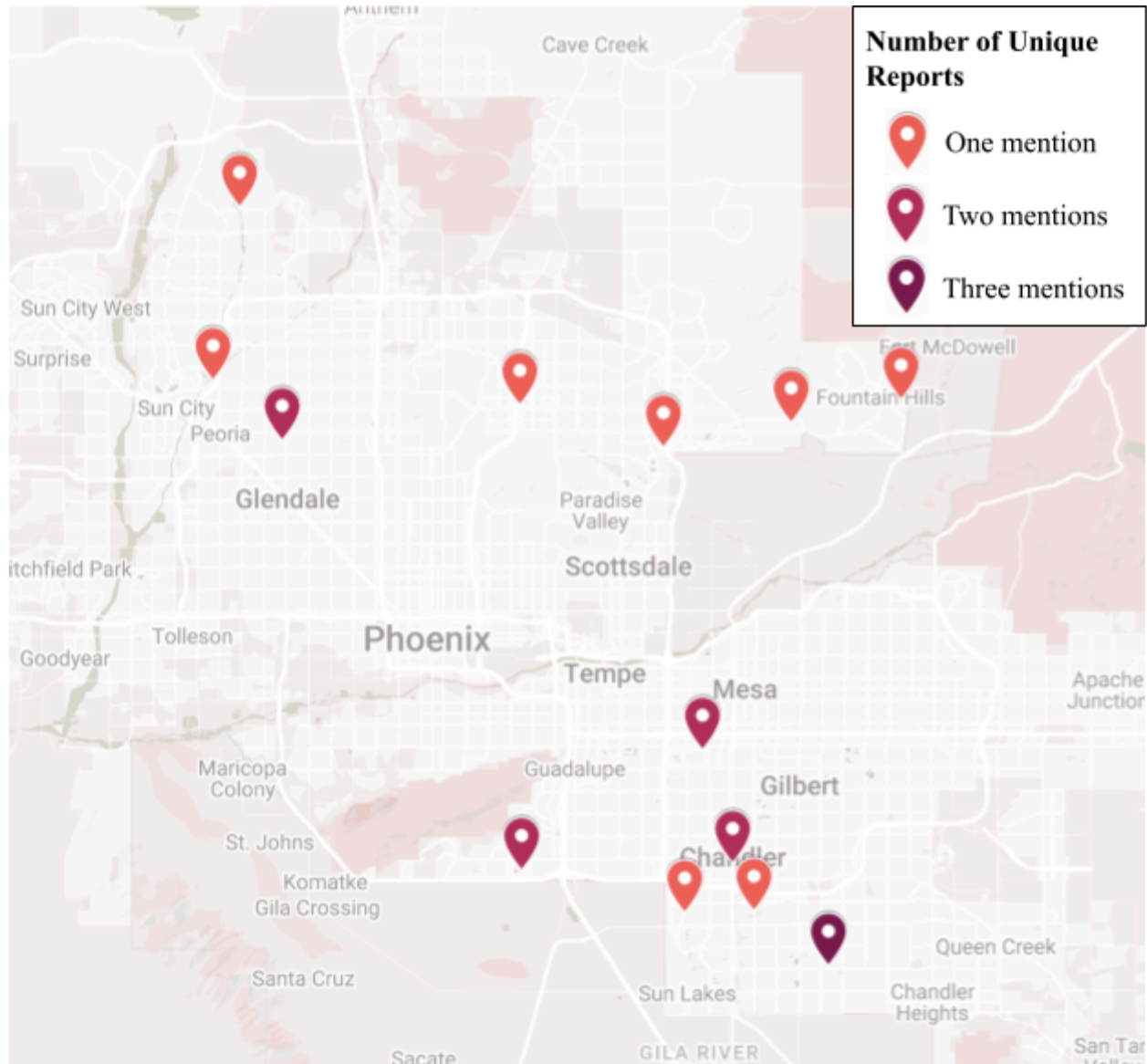


Figure 4. Map of reported “favorite parks”.

4. Survey Results and Discussion



Figure 5. Veteran's Oasis Park Map. Veteran's Oasis Park (website).

Veteran's Oasis Park in Chandler, Arizona (Figure 5) was the most popular park (N=3). Equipped with a play area, fishing lake, hiking trails, ramadas and picnic tables, an amphitheater, and an Environmental Education Center, the park supports city and school education programs, physical activity, imaginative play, and outdoor concerts. It also has a police substation on site. Similarly, the 24 TGM who reported having a favorite park appreciated park characteristics related to convenience, comfort, passive and active recreation, and design. While proximity to home and/or school was key for most of the survey takers (N=18, 75%), safety (N=6, 25%), passive social spaces (N=9, 37.5%), exercise opportunities (N=8, 33.33%), and natural features (N=10, 41.67%) were also important to several of the respondents (Table 5). Some participants made specific comments about recreational amenities like playground structures, sport courts and

4. Survey Results and Discussion

fields, “spinny chairs”, and swing sets. The TGM like parks that offer opportunities for meaningful and interactive recreation, provide spaces for relaxation and socialization, foster community, maintain a perception of safety among visitors, and have an abundance of natural features integrated into the landscape. Though, more importantly, the TGM like parks with those features *because* they have those features.

Characteristic	Count (N=24)	Percent
It is close to my home/school.	18	75.00
I feel safe.	6	25.00
There are places to hang out/talk with friends.	9	37.50
It is a good place to exercise.	8	33.33
It has plenty of plants/nature.	10	41.67

Table 5. The count and percent of TGM who selected each *Characteristic* as a reason that they like their favorite neighborhood park.

4.2.2 What would the TGM like to change?

The participants were asked, “Are there any changes you would make to your neighborhood park?” Of the 42 respondents, 36 discussed ways that they would change their neighborhood park. The responses were coded into three categories: 1) Features, 2) Maintenance, and 3) Comfort (Table 6). They also responded to several likert-scale questions, indicating the degree to which they agreed with certain sentiments and claims about their local park (Appendix G). Summary analyses of these responses provide additional context and insight into their open-ended responses.

4. Survey Results and Discussion

Categories	Count
Features	28
Amenities	16
Landscape and Design	15
Recreation	13
Maintenance	8
Repairs	1
Cleanliness	7
Bathroom Cleanliness	3
Comfort	9
Safety	3
Climate	8

Table 6. Counts of comments regarding desired park changes by coded categories and subcategories.

Features – Amenities, Recreation, and Landscapes. Most of the respondents (N=28) mentioned a desire to change one or more of their neighborhood’s park features (Table 6). Many of the comments alluded specifically to the park’s *amenities* (N=16), requesting more tables and places to sit, sidewalks, vending machines, and bathrooms or significant improvements to the park’s bathrooms. Respondents also desired improvements to their park’s *landscape and design* features with some requesting less sand and others hoping for a garden or trees for a scenic walk for some shade. Some respondents specified that they would like to have specific *recreational features* added to their local park (N=13). They requested more playground equipment, sport courts and fields, zip lines, “activities for older kids”, and swings — several girls wanted swings, or more swings. Structured community activities may also be important aspects of improving the TGM’s recreational outlook. Those who reported that their parks hosted community events and

4. Survey Results and Discussion

classes also tended to agree more that there is plenty for teens to do at their neighborhood park than those whose parks did not (Appendix F.6). This could be explained by the possibility that a teenager who is engaged with the events taking place at their neighborhood park may also be more aware of the breadth of activities available for teens at their park than a teenager who is not paying as much attention. Alternatively, hosting regular community events and structured activities at neighborhood parks might draw more teenage visitors, more often.

Maintenance. Seven of the respondents said that they would like to see improvements in the park's maintenance (Table 6). They mentioned poor upkeep and weeds, but, overwhelmingly, they wanted cleaner, better monitored restrooms. In fact, a fourth of the respondents who commented on changes they would like to see in their neighborhood park (N=9) made a comment about the restrooms. Other questions asked in the survey allude to poor restroom maintenance across the board. Only 15.38% (N=6) of the survey takers reported having access to clean bathrooms at their neighborhood park (Appendix G.2). Neighborhood park restrooms in the Phoenix metro area also seem to lag in terms of privacy, with 28.2% of respondents somewhat or strongly disagreeing with the statement, "The restrooms provide me with enough privacy" (Appendix G.2). Simple enhancements to park maintenance as well as the privacy and hygiene levels of park restrooms will not necessarily pull more teen girls and gender minorities to that park, but such enhancements stand to eliminate the deterring effect of discomfort that the presence of such minutiae cause.

Comfort – Safety and Climate. Many of the TGM also desired improvements related to park comfort. These comments were split into two key elements: Safety and Climate (Table 6). With respect to safety, several of the respondents indicated discomfort with the presence of

4. Survey Results and Discussion

unhoused people or loitering in general. One girl wrote about wanting more neighborhood supervision of the park, claiming “the one park i have near my house is not very safe, people often idle in the parking lot at night and watch those who play at the park. many of us feel unsafe.” Safety concerns were shared by many of the survey takers. Only 32.35% of respondents reported having “Good Lighting” in their neighborhood park. Over 35% of respondents indicated not feeling comfortable at the park alone; 43.59% (N=17) indicated not feeling comfortable at the park at night; and, more than half of respondents (51.29%) agreed that they sometimes felt intimidated by groups of other teens (Appendix G.2). However, most TGM reported feeling generally safe in their neighborhood park (Appendix G.2). A chi-squared hypothesis test revealed that there was a statistically significant relationship between the Girl Scouts in the sample having what they perceived to be “good lighting” in their neighborhood park and feeling comfortable in their neighborhood park at night ($Pr = 0.018$). This suggests that design features that enhance visibility may positively impact young women’s perception of nighttime park safety.

Characteristic	Count (N=39)	Percent
There were more shade.	27	69.23
There was a water feature like a pool/splash pad.	18	46.15
I lived somewhere with cooler summer temperatures.	28	71.80

Table 7. The count and percent of TGM who selected that they “Strongly Agree” or “Somewhat Agree” with the claim that they would visit the park more in the summer if the statements listed were true.

4. Survey Results and Discussion

Aside from safety, many girls wished that their neighborhood parks would be more comfortable on hot days. Five of the participants wanted more trees or shade in general to help them escape the sun and prevent sunburns. One participant requested misters or fans for hot days and another desired greater access to drinkable water. Outside of the group of Girl Scouts who specifically requested climate relief in their parks, the majority of survey takers agreed that features and amenities that promoted heat relief would make them more likely to visit their neighborhood park during the summer. More than two-thirds of respondents agreed that they would visit the park more in the summer if there were more shade, and a similar fraction said that they would do the same if their neighborhood park had a water feature like a pool or splash pad (Table 7). Nearly half of the respondents said that they would visit the park more in the summer if there were indoor facilities to help them escape the heat (Table 7). The alignment between the TGM's desire for heat relief in their neighborhood parks and previous results regarding the degree to which extreme heat deters the TGM from visiting the park (See 4.1.3 Summer Heat and Neighborhood Park Use) suggests that imposing climate-related changes to neighborhood parks could be a highly effective method for motivating year-round neighborhood park visitation among their demographic.

4.3 Survey Discussion

I found that park visitation among TGM drops off dramatically from childhood to adolescence (See 4.1.1 Visitation Frequency). As reported in Table 2, 93% of TGM in the sample reported visiting the park frequently or occasionally as a child. Conversely, 4.76% of respondents reported visiting the park frequently as teens, 33.33% reported doing so occasionally, 42.86% rarely, and 19.05% never (or almost never). Existing literature using direct observation

4. Survey Results and Discussion

measurements has previously found that teenagers are less likely to be in a park than children, and, when teenagers are present in a park, they are far more likely to be male (Cohen et al., 2016; Marquet et al., 2019). Many urban stakeholders have extrapolated the conclusion that park use decreases from childhood to adolescence from this research (Walker & Clark, 2023), however, a myriad of other time-varying factors could explain why there are more children than teens in parks today. Even though my results are based on self-reported data, they are novel in that they measure a specific individual's reported park use frequency at two points in time: childhood and adolescence. These results directly support the conclusion that park use decreases as girls and gender minorities move through their youth.

Logistical barriers, especially time contractions related to the increasing demands of school and extracurriculars, were the most commonly cited reason for the self-perceived decrease in park visitation frequency. The TGM also discussed how changes in recreational preferences meant that the park no longer made sense as a recreational destination. Parks do not offer activities that are interesting to them as adolescents. More than half of respondents (60%) reported feeling like there was not much for teenagers to do at their local park, and 80% reported feeling like the park was more tailored to little kids than to them. Even if they could find something to do at the park, visiting the park is not their first choice when it comes to deciding how to spend leisure time. With limited time and transportation options, this means that most girls rarely, if ever, visit their local parks.

The TGM also reported how often they used various features to which they had access in their neighborhood park. A higher percentage of respondents reported using passive recreation features like swings and picnic tables more often than sport facilities like courts and fields. They

4. Survey Results and Discussion

also visit the park more so out of obligation or intentional events (e.g. organized sports, family gatherings, community programs, etc...) than for spontaneous recreation (See 4.1.1 Visitation Frequency, 4.1.2 The Nature of Adolescent Park Use). This finding should not be used to conclude that teenage girls and gender minorities largely prefer passive play to active play, or that they would not visit the park for spontaneous recreation under any circumstances. Rather, it signals that the current recreational offerings in neighborhood parks are not exciting enough to promote active recreation or motivate elective park visitation among the TGM.

As hypothesized, the TGM reported that extreme heat significantly affects the timing and frequency of their park visitation (See 4.1.3 Summer Heat and Neighborhood Park Use). The majority of respondents (70.27%) reported that they were least likely to visit the park in the summer. Extreme heat is the likely culprit, since 89.74% of the respondents reported being less likely to use the park on hot days, and 75% of respondents claimed that they would visit the park more in the summer if they “lived somewhere with cooler summer temperatures.” For the TGM who are busy with homework and extracurriculars during the school year, summertime park use may be their only option, but extreme daytime temperatures restrict visits to nighttime. Unfortunately, findings of poor safety perceptions of nighttime among the TGM imply that nighttime visits may be a non-option. Extreme heat thus further exacerbates logistical barriers to park visitation.

When prompted to share the changes they would make to their local parks, the TGM largely centered their feedback around features like amenities and recreational facilities, maintenance (specifically, with respect to the restrooms), safety, and climate comfort — nearly every aspect of neighborhood park design (See 4.2.2 What would TGM like to change?). Based

4. Survey Results and Discussion

on these results, it is reasonable to conclude that, at least in this sample, Phoenix area neighborhood parks are not meeting the needs and interests of adolescent girls and gender minorities. The depressive effect of extreme heat on TGM park visitation and the surplus of comments from the TGM requesting heat adaptive measures in their local parks demonstrates how gender sensitive design, in this case, is climate sensitive design (See 4.1.3 Summer Heat and Neighborhood Park Use, 4.2.2 What would TGM like to change?).

Designing public parks that both prioritize extreme heat mitigation and meet the needs and interests of these teenage girls and gender minorities is a matter of seriously considering their feedback about various aspects of public park design. In this section, the TGM shared general feedback about the parks with which they had personal experiences. Since these experiences are personal, the feelings that the TGM have about neighborhood parks are highly specific and limited to the neighborhood parks to which they have access. To develop more detailed and prescriptive recommendations for gender and climate sensitive park design, I control for the neighborhood park in the next chapter by analyzing the results from the workshop at Hudson Park (Parque de Soza) in Tempe, Arizona.

Workshop Results and Discussion

In this chapter, I discuss the results from the workshop at Hudson Park (Parque de Soza). The workshop acted as a site-specific assessment to thoroughly evaluate the quality of a Phoenix metro area neighborhood park as well as determine how the TGM would feel about the same neighborhood park in the aggregate.

Hudson Park, recently renamed to Parque de Soza, is a neighborhood park located in the suburban area of Tempe, Arizona, at Cedar St. and Spence Ave. It features a basketball court (Figure 6), volleyball court (Figure 6), skatepark (Figure 7), detached restrooms (Figure 8), ramadas with picnic tables (Figure 8), a playground (Figure 11), a splash pad, and a public art installation called *Peace Labyrinth*, which functions as an amphitheater (Figure 19). There is also a large grassy field in the center of the park, a concrete walking path around the perimeter, and several trees and plants adorn the landscape.



Figure 6. Sport courts at Hudson Park (Parque de Soza): Volleyball Court (left), Basketball Court (right), captured March 16th, 2025.

5. Workshop Results and Discussion



Figure 7. Hudson Park (Parque de Soza) skatepark, captured March 16th, 2025



Figure 8. Amenities at Hudson Park (Parque de Soza). Detached bathrooms with water fountain in front (left) and ramada providing shade for picnic tables (right).

5.1 Observations Template

5. Workshop Results and Discussion

Overall, the TGM were intrigued by Hudson Park's design and aesthetic quality but identified room for improvement. When asked to share their first impressions of Hudson Park, some groups described it as seemingly safe and inviting, but other groups were concerned about the condition of the bathrooms and various hazards in and around the park. The TGM also found Hudson Park to be relatively well maintained and identified heat relief measures like shade and water stations throughout the park. Lastly, there was a general consensus among the groups that Hudson Park was limited in the activities it supported for teenagers, concluding that there is not much for them to do there.

5.1.1 Design and Aesthetic

In terms of Hudson Park's design and aesthetic, the TGM largely felt that it was artistically attractive, colorful, and inviting. Though, some found the park's design and aesthetic to be a bit lackluster.

When asked to share their first impressions of the park, several of the groups noted the design and aesthetic qualities of the park. Several groups mentioned the landscape, making note of greenness/greenery/green spaces (x5), a large rock, and open spaces.



Figure 9. Examples of vegetation in Hudson Park (Parque de Soza), captured on March 16th, 2025.

5. Workshop Results and Discussion



Figure 10. Hudson Park (Parque de Soza) field as an example of open space, captured on March 16th, 2025.

Two groups mentioned the “nice”, “colorful” playground equipment.



Figure 11. Colorful play structures at Hudson Park (Parque de Soza), captured March 16th, 2025.

5. Workshop Results and Discussion

Three groups mentioned that the amphitheater and sculptures inside of it caught their eye (Figure 19). Others were fond of the public art work (Figure 12), and the space was more generally described as being “bright” and “active and large.”

The groups were also asked to describe the aesthetic quality of the park by checking whether certain aesthetic features were present. Five of the eight groups observed murals/paintings in the park, and a different combination of, but still five of the eight, groups observed sidewalk or pathway art (Figure 12). Seven of the eight groups checked the box stating “Playground equipment is colorful and attractive,” and six of the eight groups checked the box indicating the presence of “Interesting seating areas.” The groups were split when it came to the park’s instagram-ability with exactly half of the groups indicating that the park “would be a good place to take photos for social media.” Six of the eight groups also agreed that the park has a “creative/pretty landscape design.” When asked to estimate “what fraction of the park is covered in natural features like trees, grass, and plants,” the average group response was just over $\frac{2}{3}$ ’s, with a standard deviation of 0.0938.

Each group also shared the parts of the park that stood out to them as focal points. Many groups mentioned sculptures and artwork integrated into functional aspects of the space like sculptures and brick amphitheater. Four groups mentioned the amphitheater specifically. Several activity areas were called out, including the swings, playground (x3), water park, skate park (x5), volleyball court, tire swing, basketball court, and teeter totters.

Not every observation was positive. A few groups mentioned that the park looked older, and the equipment appeared outdated. One group stated that the park looks “less appealing the more you walk around.”

5. Workshop Results and Discussion



Figure 12. Examples of sidewalk/pathway art and sculptures in Hudson Park (Parque de Soza), captured March 16th, 2025.

5.1.2 Comfort (Safety)

The groups expressed mixed opinions of Hudson Park's safety in their responses. While a

5. Workshop Results and Discussion

few groups noted a “friendly vibe”, “lights for night”, an absence of unhoused people, and that the park seemed “safe” in their first impressions, other groups felt differently. Two groups brought up safety hazards like someone banging their head on the rock wall, no marked crosswalks in the street, and missing safety rules in the skate park. Half of the groups took issue with the bathrooms in their first impressions, describing them as unsafe and not private due to the absence of doors on the bathrooms (Figure 13). Others felt that the lighting design failed to maximize safety as some lights could be turned on and off with a button and the volleyball court lights seem to not work all day. One group did notice and express concern with the presence of unhoused people in the park, and another, to summarize their safety concerns, stated that they “would not come [to the park] alone.”



Figure 13. Women’s restroom at Hudson Park (Parque de Soza), captured March 16th, 2025. The gate to the restroom pictured (left) is locked in an open position during the park’s opening hours.

With respect to *Comfort (Safety)*, the groups were also asked to report whether certain

5. Workshop Results and Discussion

safety features were present in the park. The park scored poorly for perceived security with only two of the eight groups affirming the presence of fences/gates around the park and none of the groups affirming the presence of locks/doors on the bathrooms.

Some literature suggests that urban design interventions that enhance visibility may positively influence women's perceived sense of safety in public space (Navarrete-Hernandez et al., 2021) Hudson Park scored high for perceived visibility and exposure with six of the eight groups reporting that there are few if any "corners or places for someone to hide" and all eight groups reporting that "the street is visible from any part of the park."

Hudson Park also scored moderately in terms of perceived emergency safety, with seven of the eight groups reporting that there are "spaces to enter or exit the park on all sides" and "crosswalks, sidewalks, and/or paths that allow someone to enter/exit/walk around the park safely at all times." The groups were prompted to rate their perceived ability to obtain help in the case of an emergency (on a scale of 1 to 5):

Rating out of 5	1	2	3	4	5
Number of Groups	0	2	2	2	2

When asked to explain their ratings, the groups provided a range of explanations. Their comments were coded into three categories: Visibility & Exposure, Ability to Contact Help, and Capacity for Access to Emergency Services. Some comments regarding Visibility & Exposure were positive, alluding to the park's open design and proximity to homes. Others saw the park's Visibility & Exposure as lacking or even problematic in it of itself, pointing out the fact that "anyone could walk up to you." In terms of Ability to Contact Help, three of the groups were

5. Workshop Results and Discussion

largely critical, voicing concerns about the absence of an emergency call button/box, the lack of people in the park, and a perception that neighbors would not respond to calls for help. Lastly, two groups commented on Capacity for Access to Emergency Services, with one group conceding the presence of side streets but no easy access to emergency services and another perceiving the park as easy for cars to access.

Overall, the TGM perceived the open design of the park to be a positive attribute of the park's safety. They expressed concerns about poor lighting design in some areas in the park, and they took significant issue with missing bathroom doors, amplifying safety and privacy concerns. Many groups also noted that there were not many other people in the park, indicating that community surveillance may be an essential element of promoting perceived safety. This aligns with existing literature surrounding the concept of "eyes on the street" and how the natural surveillance of space by residents, shopkeepers, and other patrons of the space are critical in helping young users feel comfortable (Jamme et al., 2021).

5.1.3 Comfort (Maintenance and Environment)

The attendees were generally displeased with the park's maintenance and environment. Groups took note of cigarette butts on the grass, dead plants, and broken equipment. A couple groups also wanted more benches and felt like the current benches were not size inclusive, as they either had dividers or were too tall (Figure 14). Although, their comments were not all bad. Several of the groups appreciated the park's cleanliness, including that of the bathrooms despite their issues with the missing doors. They also observed the presence of shade for environmental comfort (Figure 8, right).

5. Workshop Results and Discussion



Figure 14. Bench with dividers at Hudson Park (Parque de Soza), captured March 16th, 2025.

With respect to *Comfort (Maintenance and Environment)*, the groups were also asked to report whether the park was well maintained and sporting a relatively comfortable environment. The park scored relatively well with seven of the eight groups affirming that “most garbage is properly disposed of” and five of the eight groups affirming that “the bathrooms are clean and hygienic.” Attendees also unanimously agreed that the park was generally well taken care of and was not overwhelmed with social minutiae like “off-putting graffiti or vandalism” (Jamme et al., 2021).

Assessing the extent to which the park promoted a peaceful environment for socialization and relaxation, seven of the eight groups observed the presence of “benches/places to rest all around the park” and “private places to socialize that are away from active areas like the playground or basketball courts.” Though, only five of the eight groups felt like there were “quiet/peaceful places to escape noise (from traffic or people).”

Hudson Park also achieved a fair score for its provision of climatic comfort, with five of the eight groups indicating that there is an “abundance of shade (Artificial – ramadas and

5. Workshop Results and Discussion

canopies – or Natural – trees and bushes)” and there are “water fountains or places to easily access water.”

Hudson Park was relatively well maintained, and the TGM appreciated the extent to which the park was designed to mitigate heat impact for its patrons. Though, the park still has room for growth in terms of being an well-kept inviting space for teenage girls and gender minorities. Moreover, parks located in hot desert cities should be held to higher standards when it comes to designing for heat relief, considering the earlier finding that heat can make the park virtually inaccessible during the summer. Thus, Hudson Park, and other parks in the Phoenix-metro area should act liberally when it comes to allocating shade coverage and water access for visitors.

5.1.4 Activities

Across the board, the TGM offered constructive criticism as to the activities offered by Hudson Park. They noted that the playground looked nice and that there appeared to be a lot of things to do at the park. Though, they mentioned that the playground seemed centered toward children, and, even though the swings looked interesting to them, they were not sized appropriately for older bodies. One group also mentioned that the only activities and equipment that served teens were sports-related.

When asked to imagine activities they saw themselves doing in the park, many groups mentioned swinging on the swings, playing volleyball, basketball, or soccer, using the skate park, playing on the playground, and having a picnic. When asked to imagine activities that someone else would do, they imagined teen boys and older teens using the skatepark; “tall people” and teenagers playing basketball; little kids playing on the playground; families having

5. Workshop Results and Discussion

picnics/parties; as well as older people relaxing and doing yoga. Overall, the TGM recognized certain activities as more suited to others than themselves. For example, they imagine “tall people” playing basketball, boys using the skatepark, and little kids using the playground. It is reasonable to conclude that the TGM perceive several youth-centered activities in the park as not necessarily geared toward their gender or age.

5.2 Designing Solutions

I analyzed the solutions designed by the TGM at the end of the workshop. Of the 55 workshop attendees, 30 shared their solution maps with me. The majority of the annotated park maps were detailed and specific, demonstrating significant thought and effort from the TGM. Continuing the Hudson Park case study through the lens of the Public Space Index, I analyzed the solution maps and coded suggested changes and features into the four solutions categories that mirror the Observation Template: Design and Aesthetic Insights, Comfort (Safety) Requests, Comfort (Maintenance and Environment) Solutions, Activity Suggestions.

5.2.1 Design and Aesthetic Insights

Suggested park solutions that were deemed to visually alter or significantly change the layout of Hudson Park, were coded as Design and Aesthetic Insights. Of the 30 solution maps, 24 included changes that fit into this category. Three subcategories of changes stood out, in particular: vegetation, art, and walking paths. More than half (N=16) of the solution maps submitted included more trees, plants, and flowers throughout the park for both aesthetic and shading purposes. Five of the solution maps specified the need for a walking path through or around the park to improve connectivity between park features. Increasing the paved

5. Workshop Results and Discussion

connectivity of public parks may also streamline access for people with disabilities that impair movement. While the TGM were not asked to provide justification for their recommendations, paths may also enhance one's sense of belonging in public space as walking on a dedicated path rather than cutting through open space – and potentially *interrupting* other activity – may empower TGM in their use of the space. Lastly, one third (N=10) of the solution maps recommended the inclusion of more public art like murals and sculptures. Many of the suggestions for murals were targeted in recreation areas like the skate park, playground, and hangout areas (Appendix H). This suggests that public spaces' aesthetic character is important to the TGM, and a space's "Instagrammability" should be considered in its design (Shannon, 2019).

5.2.2 Comfort (Safety) Requests

Features of solution maps that were included with the apparent intention to enhance the physical or psychological safety of the park and its patrons were classified as Comfort (Safety) Requests. Just over half (N=16) of the solution maps included features in this category.

On trend with earlier findings (See 5.1 Observations Template), requests regarding the park's restrooms design were a focal point of the solution maps. Out of the 16 safety related suggestions, 12 were related to the bathrooms, eight of which specifically requested the installation of doors on the restrooms. The lack of doors on the restrooms is largely a relic of antiquated architectural guidelines, when it was once thought that the absence of doors promoted safety by making monitoring easier. However, for women and gender non-conforming individuals, large gaps in bathroom stall doors – or, in this case, no doors at all – invoke feelings of surveillance and make people feel unsafe (Lewkowitz & Gilliland, 2025). While ease of cleaning and maintenance is also a cited reason for this practice, bathroom cleanliness concerns

5. Workshop Results and Discussion

issued by several of the TGM in the case study seem to imply that bathroom maintenance is lagging regardless. Three of the submissions also included requests for gender-neutral or non-binary friendly bathrooms, aligning with a larger literature on the insecurity of gender non-conforming people when using public restrooms (Lane et al., 2021). While a bathroom door may not necessarily make people physically safer, its potential to alleviate young women and gender minorities' perceived safety concerns in public space is essential for reducing barriers to park use.

Some TGM called for other security measures. Over half (N=9) of the submissions that featured safety enhancements did so in the form of improved lighting. Lighting requests came in several forms, with one specifying the need for improved lighting in specific recreational areas (e.g. playground and volleyball court) and another proposing a perimeter of lighting all around the park. A few of the suggestions were quite insightful with one clarifying that in executing their desire for increased lighting, one should be careful not to disturb the animals. Another was creative in suggesting that function could be integrated into form by turning a centerpiece rose pole in the amphitheater into a light. Again, lighting proves important to the TGM's perception of park safety. While better lighting may not make teenage girls and gender minorities actually safer, like bathroom doors, improving perceptions of visibility and awareness can make the park more accessible to TGM outside of daylight hours.

Aside from bathroom privacy and lighting, three submissions suggested the installation of gates and fences, the presence of crosswalks, and the addition of rules for the skatepark. Meeting the safety desires of teenage girls and gender minorities, regardless of those changes' capacity to actually enhance park safety, is essential in reducing barriers to park use. Bathroom privacy,

lighting improvements, crosswalks, and fences all may enhance perceived safety in neighborhood parks and thus boost visitation among members of this demographic.

5.2.3 Comfort (Maintenance and Environment) Solutions

More than two-thirds (N=23) of the 30 solution map submissions included feature changes that were categorized as Comfort (Maintenance and Environment) Solutions. Feature changes in this category are those that enhance the park's maintenance, improve patrons' capacity to cope with the climate while using the park, or enable physical or psychological respite from typical park activities. The most common suggestions were related to cooling and seating.

More than half of the 22 maps that were coded into this category (N=13) incorporated heat relief measures like shade features or indoor access. Five TGM included a ramada in their design, and eight submissions added shade or coverage of benches and recreational areas, more generally. A few specifically requested shade cover in the form of trees, vegetation, or recycled cotton. Given the intense heat profile of the Phoenix-metro area, it is natural that addressing heat concerns would be a priority for all demographic groups. In line with these heat concerns, a few (N=5) TGM incorporated water and drink stations in their designs, desiring convenient water access for heat adaptation.

Benches were included as an addition to Hudson Park in 12 of the solution maps. General benches, picnic tables, and ramadas were coded as Comfort (Maintenance and Environment) Solutions, and dedicated hangout, relaxation, and lounge areas were classified as Activity Suggestions, as sitting on a bench oriented toward an activity is more so a break from park activities rather than an activity itself.

5.2.4 Activity Suggestions

Nearly every solution map (N=28) included suggestions for activity related features. The Activity Suggestions category encompasses any changes that add or remove existing activity infrastructure, enhance the enjoyability of existing activity infrastructure, or otherwise enable the performance of park activities.

The top suggestions were thoughtfully designed lounge and hangout areas (N=17) and swings: *more* swings and *bigger* swings (N=17). With respect to hangout areas, the TGM mostly desired structures like hammocks (N=8) and treehouses (N=5) in secluded areas, surrounded by vegetation, art, and cozy decorations. Many of the suggestions were visibly inspired by reference photos provided to TGM during the Hudson Park workshop (Figure 15). These areas were described as places to lounge (N=3), read (N=4), relax (N=1), and hang out (N=2). The desire among TGM for comfortable, dedicated spaces where they can passively hang out or just “be” aligns with a myriad of social science research dedicated to understanding how teenage girls and gender non-conforming people use space. Such research has come to understand “hanging-out” and other forms of passive recreation as an act of playful engagement and claiming spaces (Pyyry, 2016; Hjort & Larson, 2025).

As play equipment, swings are unique in the sense that they enable both active – swinging – and passive – talking to friends or observing surroundings – recreation at the same time. Naturally, they were extremely popular among the TGM. This finding is consistent with existing research consensus that swings, specifically those designed for teenage use, are incredibly popular among this demographic. Some TGM (N=5) simply advocated for more swings, but many were quite specific with their desires, drawing on reference photos

5. Workshop Results and Discussion

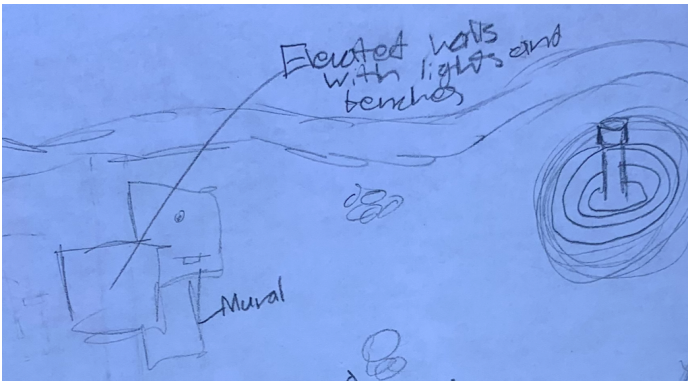
Workshop Reference	Solution Map Example
	
	
	

Figure 15. Examples of features on solution maps that correspond to seating and hangout area inspiration images presented at the Hudson Park workshop. See Appendix D.6. for image credits.

5. Workshop Results and Discussion

presented at the workshop (Figure 16) as well as their imaginations. Four of the drawings advocated for the *Circle Swings* from the Umea Case Study featured in Figure 16 (top), and another four referred to the *Hang Time* swings from Figure 16 (bottom). A few others advocated for tandem or group swings, highlighting the TGM's desire for swings as a medium for socialization. There were also a few comments about larger, taller swings that were built to fit the bodies of older children. One map even featured a swing designed to accommodate a wheelchair user. Overall, the TGM want swings that are fun, visually appealing, conducive to socialization, and designed to fit their bodies.

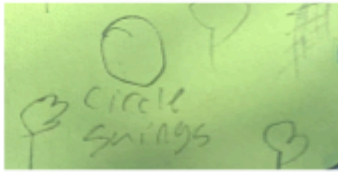
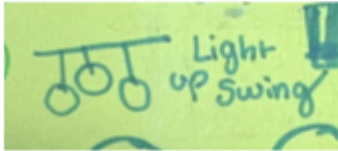
The desire for interactive play structures was not limited to swings, climbing structures (N=8), obstacle/ropes courses (N=7), and ziplines (N=5) commonly occurred in the solution maps. Climbing structures may be popular among teenage girls beyond the scope of this research per the Rosens Roda Matta case study. When girls in Malmo, Sweden aged 16-24 were consulted for the park's design, a climbing wall was a key feature of the final product (Make Space for Girls, Figure 17). These play structures could be incorporated into playgrounds designed for adolescents and older children.

Several of the suggested activity amenities (N=11) were sport-related. Some of the suggestions were typical park amenities like soccer nets (N=6) and baseball fields (N=2), but others were not typically included in neighborhood parks. For example, two of the TGM included a Gaga Ball pit in their drawings, two added workout areas, and two more drew bike activity areas on their maps. Four of the submissions had equipment areas, mostly near the basketball court, for public use. Having sport equipment accessible to the public may encourage non-typical, like teenage girls and gender minorities, to try new activities at the park. Six of the

5. Workshop Results and Discussion

submissions also suggested that certain sport features be removed or reduced in size; the skate park (N=3) and volleyball court (N=2) were most common. This indicates that certain recreational amenities may act as deterrents to neighborhood park use among these individuals, but they may also just prefer that limited park space is allocated toward different features.

Figure 16. Examples of features on solution maps that correspond to swing-related reference photos from the Hudson Park workshop.

Workshop Reference	Solution Map Examples
	
	
	
	
	

Note. From *Frizon by Tyrens* [Photograph], by Andreas Nilsson, 2017, Interior Design (<https://interiordesign.net/designwire/swedish-design-moves-new-york-spotlights-democratic-architecture/>); From *Teen Girls Need Better Public Spaces to Hang Out* [Photograph], by Höweler + Yoon Architecture, 2021, Bloomberg City Lab (<https://www.bloomberg.com/news/features/2021-05-28/we-need-more-public-space-for-teen-girls>)

5. Workshop Results and Discussion

Lastly, leaning into a more unstructured and imaginative form of play, eight of the solution maps had a dedicated space for a stage, performance area, or theater. This suggestion is especially pertinent to the gendered aspect of this discussion, given that teen girls are far more likely to participate in performing arts activities (e.g. dance, drama, music lessons, etc...) and value the arts than their male counterparts (Smyth, 2016, 2020).



Figure 17. Shaikly, A., & Mella Lira, B. (2022). *Rosens Roda Matta, Malmö, Sweden* [Photograph], by Bethany Francis, 2020 (<https://doi-org.ezproxy1.lib.asu.edu/10.1080/13574809.2022.2125370>)

In summary, the personal park preferences and concerns that the TGM outlined in the survey persisted in their analysis of Hudson Park. They wanted a more aesthetic park that featured vegetation and art. They wanted lights and basic privacy features like lockable doors on bathrooms and gender-inclusive restrooms. They also desired enhanced climate comfort and wanted more opportunities for imaginative and interactive play. While the Solutions Design

5. Workshop Results and Discussion

activity was specific to Hudson Park, the suggestions made in the solution maps were quite general and provide critical insight into the parts of neighborhood parks that are and are not working for the TGM. For example, it is likely that TGM who are uncomfortable with the absence of bathroom doors at Hudson Park would prefer to have lockable, closed doors on bathrooms in other neighborhood parks. Likewise, the overwhelming desire for the addition of swings and interesting social hangout spaces to Hudson Park indicates that other neighborhood parks could incorporate these features to attract visitors within their demographic.

Conclusions and Recommendations

The TGM have explained the nature and frequency of their neighborhood park use; described their personal experiences with neighborhood parks; outlined changes they would like to see made to their local park; worked in groups to gather data regarding the quality of a specific neighborhood park (Hudson Park); and visually defined their ideal version of Hudson Park, providing keen insight into how park planners can design parks that are amenable to the needs and interests of teenage girls and gender minorities. Furthermore, the TGM's responses to the survey and workshop activities have advanced my understanding of the relationship between park use and climate among teenage girls and gender minorities.

In this section, I deliver design and planning recommendations for stakeholders who are concerned by both climate and gender equity. These recommendations will empower gender and age inclusive park design in a way that allows planners to continue to prioritize climate adaptation in geographies like that of the Phoenix metro area with high heat-related climate risk.

6.1 Conclusions

Through this research initiative, I aimed to answer several questions about how adolescent girls and gender minorities engage with neighborhood parks in the Phoenix metro area and how park planners can more effectively design for this demographic in hot desert cities like Phoenix.

6. Conclusions and Recommendations

My first question asks how adolescent girls and gender minorities use, perceive, and feel about their neighborhood parks in the Phoenix metro area with two sub-questions: 1) how do these feelings and interactions evolve from childhood to adolescence, and 2) how does Phoenix's hot desert climate affect the neighborhood park use of teenage girls and gender minorities? Findings from the survey suggest that park use among teenage girls and gender minorities is infrequent (See 4.1.1 Visitation Frequency), and visits are generally obligatory in nature or characterized by passive recreation activity (See 4.12 The Nature of Adolescent Park Use). Moreover, the survey results support the common notion in the literature that park visitation among girls and gender minorities drops off significantly from childhood to adolescence (See 4.1.1 Visitation Frequency; Yorkshire Sport Foundation, 2022; Make Space For Girls, 2023). The nature of park use also appears to change over time as far fewer TGM reported visiting the park for elective recreation as adolescents than those who reported doing so as children.

Existing literature argues that these changes in park use are a product of the fact that adolescent needs and interests are hardly considered in public park design, and, when amenities and programming are geared towards adolescents, they mainly cater to traditionally male preferences (Yorkshire Sport Foundation, 2022; Make Space For Girls, 2023). This is largely consistent with my findings, as the second most common reason that the TGM cited for decreased park visitation in adolescence was that parks simply were not fun anymore. They felt like there was not much for them to do at their neighborhood parks (See 4.1.1 Visitation Frequency). Although, the most common reason the TGM cited for not visiting the park as often as when they were children had less to do with gender and more to do with a nearly universal

6. Conclusions and Recommendations

adolescent experience: logistical constraints (See 4.1.1 Visitation Frequency). These constraints ranged from being too busy to visit the park to not having access to transportation.

Despite being the most commonly cited reason for decreased neighborhood park use in adolescence, time constraints are somewhat sparse in the literature that explores barriers to teenage girls' and gender minorities' park use (Rivera et al., 2022; Yorkshire Sport Foundation 2022; Make Space for Girls, 2023; Hjort & Larson, 2025). One possible explanation is that research focused on the gendered aspects of adolescent park use may not center their discussions of park use barriers around obstacles that are seemingly gender neutral, like time. Though, the role of time constraints as a barrier to park use is also absent from adolescent park use research that targets all genders (Rivera et al., 2022). On the other hand, the role of proximity and transportation is discussed in the literature at length. In a systematic review of barriers to public space use among teenage girls, Hjort & Listerborn (2025) conclude that female teenagers are far more likely to visit parks and recreational spaces that are within walking or biking distance.

The TGM also expressed that they perceive their neighborhood parks as unsafe. Psychological and physical safety concerns were a common theme in the findings of both the survey and workshop, and the literature heavily emphasizes the role of these concerns in depressing park visitation among teenage girls and gender minorities (Condon et al., 2007; Polko & Kimic, 2022; Hjort & Listerborn, 2025). They largely noted privacy and restroom cleanliness as well as the effect of extreme heat on physiological safety and comfort as key safety-related issues. In both the survey and the workshop, the TGM consistently requested enhanced safety and heat-mitigation measures (See 4.2.2 What would the TGM like to change?, 5.2.2 Comfort (Safety) Requests).

6. Conclusions and Recommendations

Safety concerns also add a gendered dimension to the effect of logistical barriers and extreme heat on park visitation. That is, logistical barriers are not completely gender neutral, since the public space use of teenage girls and gender minorities is temporally limited by nighttime safety concerns (See 4.2 Motivating Neighborhood Park Visitation). This sheds light on the second sub-question, referring to the effect that Phoenix's hot desert climate has on the park use of teenage girls and gender minorities. I found that the combination of logistical barriers and extreme summer heat in Phoenix exacerbates the effects that nighttime safety concerns have on park visitation frequency (See 4.1.3 Summer Heat and Neighborhood Park Use). Moreover, extreme heat seemed to dramatically shape the TGM's park use, dictating their decision to visit the park and the times at which they visit the park. While the Yorkshire Sport Foundation (2022) reported that weather is a deterrent to park visitation, especially among teenage girls, they only mentioned *cold* and *rain* as deterring weather events and did not address the role of extreme heat. This is logical, given that the Yorkshire Sport Foundation and its audience are based in the United Kingdom, but it also highlights the importance of conducting gender sensitive park planning research in hot desert contexts.

The survey and workshop results, overall, imply that the park use of teenage girls and gender minorities in the Phoenix metro area is infrequent, passive, and obligatory. The TGM visit parks less often than when they were children and more so out of obligation than elective enjoyment because they have less time, are limited by transportation access, and take little interest in the social and recreational amenities that their neighborhood parks have to offer. They also feel that their neighborhood parks are not safe enough, clean enough, or sufficiently heat

6. Conclusions and Recommendations

resilient. Lastly, extreme heat is a significant barrier to their neighborhood park use, especially when considered in the context of poor perceptions of nighttime safety and logistical barriers.

My second question asks how adolescent girls and gender minorities perceive the quality of existing neighborhood parks in the Phoenix metro area. To answer this question, I pull from the results of the Hudson Park (Parque de Soza) workshop, which functioned as a site-specific assessment of the Phoenix metro area park. Results from the observations activity (See 5.1 Observations Template) indicate that the TGM found Hudson Park's design and aesthetic to be of moderate quality, characterized by nature and plant life, an open layout, colorful equipment, and the presence of public art. Their perception of safety in Hudson Park was mixed, with Hudson Park scoring poorly in terms of perceived security, moderately for visibility, and moderately for perceived access to emergency support. With respect to maintenance, the TGM felt that the Phoenix metro area park had room for significant improvement. They were relatively appreciative of the heat adaptive elements of the park's design like shade and water access, but they still felt as though there was much to be desired when it came to climate comfort. In terms of activities, the TGM identified plenty of activities that could be conducted in the park but felt like few of the activities were suited to their needs and interests. The results from the solutions design activity (See 5.2 Designing Solutions) also emphasize the perception among the TGM that Phoenix metro area neighborhood parks are not meeting their needs for heat relief and interactive play opportunities.

My third and final question asks what steps urban stakeholders can take to plan and design neighborhood parks that serve teenage girls and gender minorities while also striving for climate resilience. The next section will explore practical, concrete recommendations for park

6. Conclusions and Recommendations

designers, but first, I approach this question broadly. Conclusively, boredom and logistical constraints are the primary barriers to TGM neighborhood park use. Inclusive parks are thus parks that are both interesting to teenage girls and gender minorities and easy to access. Initial steps toward making neighborhood parks more inclusive for teenage girls and gender minorities should entail the addition of infrastructure and amenities that support this demographic's preferred forms of park recreation – passive and active. When the TGM do visit the park, they describe their recreation as mostly passive, but this is not necessarily by choice. Results from Section 5.2.4 Activity Suggestions demonstrate that the TGM are interested in active recreation, but the specific active recreation amenities commonly afforded to public parks simply do not interest them. Adolescent girls and gender minorities do not necessarily prefer passive recreation to active recreation in parks; they just are not excited by the active play features to which they typically have access. Thus, it is important that park recommendations emphasize the active recreation features that the TGM report using as teenagers, the types of recreation that the TGM specifically requested in the survey (See 4.2.2 What would the TGM like to change?), and the recreational amenity suggestions they provided during the workshop (See 5.2 Designing Solutions).

Incorporating such suggestions could 1) naturally make parks more recreationally attractive to teenage girls and gender minorities, and 2) partially mitigate the depressive effect of logistical barriers. This is because the role of logistical barriers is also not entirely one of access but rather opportunity cost. Many TGM may technically have the time to visit the park, but the park *not being fun anymore* discourages park visitation with the little free time they have.

Addressing logistical barriers to park visitation is both a matter of improving temporal-spatial

6. Conclusions and Recommendations

access and designing parks that sufficiently meet the needs and interests of teenage girls and gender minorities such that they choose to visit the park over other leisure activities. A park planner or designer may not be able to fully mitigate the temporal-spatial constraints that affect adolescent park visitation in the context of designing any specific park, but regional or city planners interested in promoting adolescent park visitation may consider building or improving parks near middle and high schools for convenient access. One finding that stood out was the fact that, among the TGM who reported having a favorite park, the most frequently reported reason for denoting a park as their “favorite” was because it was close to their home or school. This supports the notion that access is at least in part a matter of convenience.

Furthermore, given the intersection of time constraints, nighttime safety concerns, and behavioral responses to extreme heat, addressing any one of these issues could potentially reduce the extent to which all three inhibit TGM park visitation. For example, enhancing a park’s actual and perceived park safety could make visiting the park at night more feasible for TGM. In turn, someone who is too busy to visit the park during the school year but put off by extreme outdoor temperatures during their summer break from school may now be able to visit their neighborhood park more often at night during summer months. Likewise, implementing measures for extreme heat mitigation and relief could promote park visitation over summer break. Specific park design recommendations that aim to enhance park safety perceptions and combat the effects of extreme heat in a gender sensitive manner are outlined in the next section.

6.2 Recommendations

A combination of factors influence park use among teenage girls and gender minorities. It is essential for planners and designers to recognize how design choices can either encourage or

6. Conclusions and Recommendations

discourage park use. Some design elements serve as push factors, creating barriers to access—such as logistical constraints, bad weather, and poorly maintained bathrooms. Conversely, pull factors attract this demographic, including social obligations (e.g., organized sports, family celebrations), scenic quality, community events, and engaging activities.

To increase park visitation among adolescent girls and gender minorities, planners should aim to reduce or eliminate push factors while enhancing both the quantity and quality of pull factors. Parks must not only meet the general needs of these groups but also motivate them to choose park visits over other leisure activities. The most effective approach is to prioritize teenage girls' and gender minorities' preferences regarding aesthetic quality, safety, maintenance, environment, and activities.

6.2.1 Incorporate more native plants and other vegetation into park design.

Incorporating native plants and other vegetation into neighborhood park design can simultaneously satisfy the aesthetic desires of the TGM while also providing heat relief. The TGM overwhelmingly reported heat as a major obstacle to park use but noted that more shade would encourage park visits on hot days: some of the only days they have free time to visit the park. Their preference for compelling natural park landscapes for both scenic and shading purposes was also a consistent theme throughout this research. Many survey respondents expressed a desire for more trees and gardens in their neighborhood park; over half of the groups in the Observations Activity highlighted landscape features; and, more than 50% of the solution map designs included trees, bushes, and gardens.

Urban tree canopies and tree-grass clusters are highly effective at reducing land surface temperatures in hot desert cities (Middel et al., 2015; Myint et al., 2015). A scenic walking path

6. Conclusions and Recommendations

lined with trees or seating areas surrounded by native plants could make parks more inviting to teenage girls and gender minorities while providing heat relief for all visitors. Implementing this approach would encourage year-round park use for these groups and improve summertime accessibility for everyone.

6.2.2 Bigger, better, shaded swings (and other active recreation amenities).

Throughout the research, the TGM made it abundantly clear that the recreational infrastructure in their neighborhood parks did not interest them, maintaining that there was not much for them to do in their neighborhood parks. Across the board, they wanted more teen-oriented opportunities for active recreation that were not centered around sport. Many TGM reported frequently using the swings and playground when they visit their neighborhood park, but they are hardly a pull factor as child-centered designs produce lackluster experiences with these features.

Of the 30 submitted solution maps, 17 indicated that their ideal version of Hudson Park would have more options for or improve current amenities for non-sport active recreation (5.2.4 Activity Suggestions). Based on these results and the TGM's survey responses, installing interactive play equipment like swings, obstacle and ropes courses, climbing structures, ziplines, and adolescent-oriented playground equipment in general may make parks more attractive to teenage girls and gender minorities (Figure 18). It is especially important to design these features with a wide range of body sizes and use cases in mind: swings should be designed to provide exciting active play experiences while also being conducive to passive socialization between friends.

6. Conclusions and Recommendations

The TGM were also quite enthusiastic about incorporating shade into these amenities to mitigate extreme heat (See 5.2.3 Comfort (Maintenance and Environment) Solutions). While trees and canvas canopies were common suggestions, one solution map featured a climbing structure with a recycled cotton roof for shade coverage. To protect against heat exhaustion and burns, planners and designers should add shade coverage to active play equipment when feasible.

Figure 18. Examples of ideal swings and interactive play structures.



Note. From *Frizon by Tyrens* [Photograph], by Andreas Nilsson, 2017, Interior Design (<https://interiordesign.net/designwire/swedish-design-moves-new-york-spotlights-democratic-architecture/>); From *Teen Girls Need Better Public Spaces to Hang Out* [Photograph], by Höweler + Yoon Architecture, 2021, Bloomberg City Lab (<https://www.bloomberg.com/news/features/2021-05-28/we-need-more-public-space-for-teen-girls>); From *Superkilen* [Photograph], by Make Space for Girls, 2021 (<https://www.makespaceforgirls.co.uk/case-studies/superkilen>); From *Renewing Yangon* [Photograph], by Doh Eain, n.d., (<https://www.doheain.com/en>)

Building entire playgrounds dedicated to adolescents may not be practical. However, planners and designers should not ignore the fact that the TGM reported using the playground well into adolescence, and they should weigh the repeated requests for larger, more challenging playground equipment that appeals to a wider range of age groups. Applying universal design

6. Conclusions and Recommendations

principles by designing equipment for a larger range of body shapes may benefit other park users like larger children and parents who want to play with their children.

6.2.3 Do not neglect the visual aspect of park features. Enhance them with public art.

Further leaning into park aesthetics, the TGM wanted to enhance the visual quality of Hudson Park (Parque de Soza) and its amenities through public art. In the Observations Activity, they were fond of the artwork present in Hudson Park. They also appreciated amenities that had artistic appeal themselves, like the labyrinth amphitheater (Figure 19) and colorful playground equipment.



Figure 19. *Rebecca Blume Rothman Tempe Public Art Director walks in the labyrinth at Hudson Park on Oct. 1, 2019 in Tempe, Ariz. [Photograph], by Rob Schumacher, 2019, The Arizona Republic (<https://www.azcentral.com/story/entertainment/events/2019/10/03/arizona-labyrinth-tempe-hudson-park/3777052002/>).*

6. Conclusions and Recommendations

The solution map submissions clearly illustrated the desire for the park's recreational elements and amenities, not just the landscape, to have an artistic quality themselves. Several of the TGM submitted designs with painted murals on the ramp in the skatepark, on the external wall of the bathroom, and near the playground. One submission included a sculpture that doubled as a climbing structure. The workshop reference photos that featured recreation equipment with artistic designs like the glowy swings of *Swing Time* (Figure 16), were among the most popular with the TGM. Simply put, they want equipment that looks cool.

With respect to climate sensitivity, many major cities have started collaborating with public artists to design climate resilient infrastructure that does not threaten but, rather, enhances the visual appeal of public space. For example, the Zeus-Eco Mural in Los Angeles, California is painted with a solar reflective coating to reduce surface temperatures on the old brick building it spans (Metropolitan Area Planning Council, 2021; Figure 20). Covering existing park structures or non-permeable surface cover with solar reflective murals can revamp the appearance of dated infrastructure while also transforming heat-absorbing materials into mediums for heat mitigation.

The City of Phoenix has also taken part in this trend, working with artists to build eye-catching, perforated transit shelters that minimize heat impact by balancing shade cover and air flow (City of Phoenix, n.d.; Figure 21). By collaborating with artists, Phoenix is able to be hyper-practical in the design of these transit shelters while maintaining the cultural character of the urban landscape (Mehrotra, 2023). Thus, public art may be a helpful tool for navigating the political economy of park planning for heat resilience by allowing planners to incorporate radical cooling infrastructure into their designs without offending the visual sensibilities of surrounding residents.

6. Conclusions and Recommendations



Figure 20. Metropolitan Area Planning Council (2021). Zeus Eco-Mural Presentation Slide. Retrieved from <https://www.mapc.org/wp-content/uploads/2021/08/081821-Cool-it-with-Art-Slide-Deck.pdf>



Figure 21. *Take the Long Way Home – Happy Valley Park and Ride* [Photograph], 2011, City of Phoenix (<https://www.phoenix.gov/arts/publicartmap>)

6.2.4 Fix the bathrooms.

The TGM took issue with the state of the bathrooms at Hudson Park and in their own neighborhood parks. In the observations activity, the TGM found the absence of bathroom doors in Hudson Park to be concerning, making the bathrooms seem unsafe and exposed. Improving the bathrooms, by adding doors or increasing levels of general cleanliness and maintenance, was one of the most common themes in the solution maps. Several TGM also requested gender neutral restrooms or single-stall all-gender restrooms. Such additions may support privacy and comfort among gender non-conforming teenagers.

The absence of bathrooms, dirty bathrooms, and poorly maintained bathrooms are barriers to park use and physical activity in parks (Yorkshire Sport Foundation, 2022). For adolescent girls and gender nonconforming people who may still be learning how to navigate their menstrual cycle, access to private and hygienic bathroom facilities are non-negotiable (Yorkshire Sport Foundation, 2022; Lewkowitz & Gilliland, 2025). To address menstruation and other hygiene-related park use barriers, bathroom stalls should have private waste receptacles and operators should consider making feminine products available in the restrooms. Neglected bathrooms are not just a minor inconvenience for teenage girls and gender minorities; they are a significant barrier to park use and physical activity in public spaces.

6.2.5 Curate cozy spaces for socialization.

Socialization should not be an afterthought in park design for the needs and interests of teenage girls and gender minorities, yet seating is often limited to a few picnic tables or benches facing other activities from the sidelines. This placement hardly centers neighborhood parks as

6. Conclusions and Recommendations

comfortable spaces for relaxation or spending time with friends, which are the preferred park activities of teenage girls and gender minorities.

The TGM desired change: well-designed hangout and seating spaces were the most common addition to the solution maps (See 5.2.4 Activity Suggestions). There should be dedicated spaces for relaxation and passive recreation throughout parks. Hammock areas (Figure 22, left), multi-level decks (Figure 22, center left), private picnic tables (Figure 22, center right), and shaded lounge-style seating (Figure 22, right) were incredibly popular additions to the solution maps.

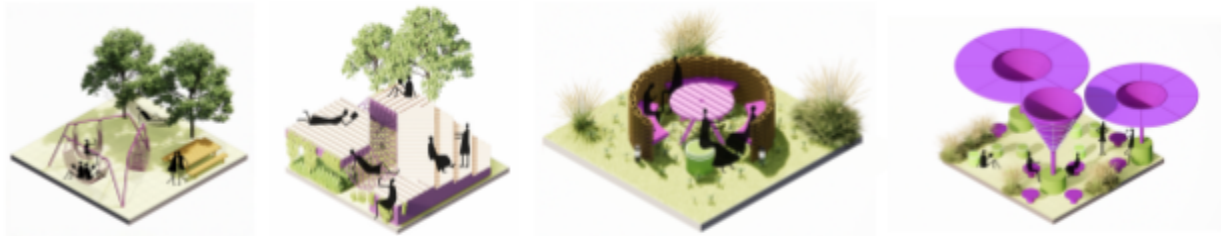


Figure 22. Seating/hangout focused workshop inspiration images from Appendix D.6.

Designing park seating areas to be comfortable, attractive, and self-contained promotes inclusivity and climate sensitivity across multiple dimensions. Aside from appealing to teenage girls and gender minorities, high quality seating areas may enhance park experiences for mobility impaired individuals like pregnant and elderly visitors, or people with physical disabilities. On hot days when active recreation is not feasible, the option to socialize in heat-adaptive seating areas can keep parks accessible to more people.

6.2.6 Focus on nighttime safety. Lighting is a must, but be strategic.

Even with dramatic design-based interventions, the capacity for design to address the effects of extreme-heat are limited in hot desert cities, where daytime temperatures can reach 120

6. Conclusions and Recommendations

degrees Fahrenheit. During the summer, there may be times where it is only safe to use the park in the early morning or at night. Unfortunately, using the park at night may be a non-option for teenage girls and gender minorities. Many of the TGM expressed in the survey that they felt unsafe or uncomfortable using the park at night. I did find, however, that there was a significant relationship with the TGM reporting that their neighborhood park has “good lighting” and feeling comfortable in that park at night. It makes sense that comprehensive lighting features were frequently integrated into the solution maps. This is consistent with other findings that better lighting may help teenage girls and gender minorities feel safer in their neighborhood parks at night (Yorkshire Sport Foundation, 2022; Bornioli et al., 2024; Hjort & Listerbon, 2025), allowing them to access parks during the hottest parts of the year.

Designers can enhance night time safety through lighting by installing shielded, downcast lighting fixtures that optimize visibility and minimize light pollution (Cokinos, 2024). Public art collaborations, as discussed in Section 6.3.3, can enable this without compromising the visual profile of parks. For example, by suggestion of one of the submitted solution maps, the rose pole in the center of the Hudson Park amphitheater (Figure 12) could also function as a light. Lighting can also be incorporated into the equipment designs (See *Swing Time* in Figure 16). Through public art collaborations, light features can both practically improve night time visibility and raise the aesthetic profile of a park.

6.3 Limitations

When choosing whether and how to implement these recommendations, stakeholders should be aware of several limitations of this research. The first is the limited sample size and potential sampling bias. After exclusions, the survey results are based off of a mere 42 responses,

6. Conclusions and Recommendations

with many questions and analyses conducted over an even smaller subset of responses. While one can reasonably draw conclusions of statistical significance from this population, it is critical to consider that these results only reflect the feelings and perceptions of a few Phoenix metro area adolescent girls and gender minorities (Etz & Arroyo, 2015). Anyone planning or designing a neighborhood park outside of this region should use this project as a blueprint to conduct their own analyses with local teenage girls and gender minorities to assess their unique needs and interests. In addition, the sample selection was limited to Girl Scouts – Arizona Cactus-Pine Council membership. As active members of an organization dedicated to the education and empowerment of young women as well as environmental stewardship, the girls and gender minorities in this sample may have stronger feelings about gender inequality and climate-related issues (Anderson & Behringer, 2010; Arneil, 2010). Hence, urban actors should consider that opinions of those in this sample may not be representative of those of the entire demographic.

Also, the true geographic distribution of the TGM is unknown, as the survey only requested county-level data and all respondents reported that they reside in Maricopa County. While the geographic distribution of reported favorite parks served as somewhat of a proxy for the TGM's residential distribution, it is neither complete nor exact. Anyone who wishes to replicate this initiative should consider recruiting a larger sample from a more diverse pool of teenage girls and gender minorities, conducting multiple workshops throughout the valley, and requesting residential zip code information rather than county-level information in the survey.

In addition, the nature of this research brought about several unavoidable limitations such that attempting to circumvent them in future projects could diminish the integrity of the research. First and most emphatically, the survey analyses are not based on direct observation but rather

6. Conclusions and Recommendations

personal recall and self-reported behavior. The TGM are asked to recall behavior and trends from almost a decade prior in some cases. Thus, there is a potential for significant error in the responses and conclusions that are drawn from them (de Nicola & Giné, 2012). Avoiding recall and self-reporting error is difficult because directly tracking individual park use for several years is not only resource and time intensive, but is also challenging in the context of research that is intended to bring about change as soon as possible.

Second, since the vast majority of the TGM were under the age of 18 at the time of data collection, privacy concerns were critical to the research design. As a result, the individual survey responses and associated demographic information were not connected to the data generated during the workshop, even though the samples were nearly identical. This restricts the potential for demographic analyses of the results from the observations template and solutions design activities. Such a decision was necessary to protect the research subjects.

Lastly, several educational interventions integrated into the workshop (See 3.5.2 Workshop Introduction, 3.5.5 Solutions Discussion) were intended to and likely did influence the results of the observations activity (See 5.1 Observations Template) and the feedback extracted from the solutions design activity (See 5.2 Designing Solutions). For example, the TGM were presented with images of potential design features from which they could draw inspiration for their solution maps. This could potentially be seen as leading, however, the goal of this research project is to design effective recommendations by directly consulting the TGM. That does not necessarily require them to operate in an informational vacuum. The TGM were also encouraged to both reflect on the aspects of the inspiration concepts that appealed to them and change aspects that did not appeal to them. Though, if a future project leader were concerned about

6. Conclusions and Recommendations

potential biases introduced through these educational interventions, they could request that the workshop participants complete the solutions design activity (See 3.5.6 Solutions Design Activity) both before and after the educational interventions are presented in the workshop.

6.4 Final Considerations

Designing parks that meet the needs and interests of teenage girls and gender minorities in a hot desert city is a matter of simply asking them what they like about their local neighborhood parks, what they would change about them, and how they would change them. The teenage girls and gender minorities in this sample wanted parks that were clean and well maintained, promoted safety at all times of day, offered interesting opportunities for meaningful socialization and non-sport active recreation, emphasized aesthetic character, and integrated extensive heat relief measures into their design.

The goal of my research was to arrive at design solutions that made neighborhood parks more conducive to the needs and interests of adolescent girls and gender minorities while also appealing to the heat-adaptive design goals of most hot desert area planners and designers. This feat was far less challenging than I initially expected because so many of the concerns, recommendations, and proposals from the TGM were centered around climate and extreme heat. Throughout the data collection process, even when the TGM were not prompted to think about extreme heat, many of them elected to include concerns and considerations related to extreme heat in their responses. One of the most important takeaways from this research is that gender sensitive design is climate sensitive design. Extreme heat negatively affects the TGM's neighborhood park use, and the TGM overwhelmingly desired improvements to climate sensitive

6. Conclusions and Recommendations

design. The assumption that gender sensitive design and climate sensitive design are incompatible in any sense is incorrect.

It is also important to remember that teenage girls and non-binary people are not a monolith. Many of the TGM did request and express interest in sport-related or stereotypically male amenities. Several of the themes also varied across age groups with older teens using different features, visiting the park at a lower frequency, and reporting an overall lower sense of park belonging than younger teens and adolescents. One's preferences and relationship with neighborhood parks continues to change throughout adolescence. Therefore, incorporating park features with the intention of targeting the needs and interests of any one group may not be efficient. Instead, planners and designers should focus on including a variety of recreational amenities that appeal to diverse interests and work for more than one group.

Hot desert cities are not a monolith either. The findings pertaining to the effects of extreme heat on park behavior and the participant suggestions for climate sensitive design interventions may be specific to the Phoenix metro area. Similar initiatives in other hot desert cities in the United States or even in Arizona might arrive at different conclusions. This is one of the first gender sensitive public park design initiatives to be executed in a hot desert city in a Global North context, so there is no other comparable literature to support or contradict my findings. Further research in other cities characterized by extreme heat is needed to verify if my findings are actually caused by extreme heat or if there are unobserved characteristics of the Phoenix metro area driving these results.

Finally, applying these recommendations will make neighborhood parks better for everyone, not just teenage girls and gender minorities. Implementing the recommendations for

6. Conclusions and Recommendations

recreation amenities may also make neighborhood parks more fun for teenage boys whose preferences do not align with the stereotypical boy their age, parents accompanying their children, or anyone who still wants to play outside after childhood. Everyone in hot desert cities would appreciate climate sensitive park design during the summer months, especially the elderly or people with disabilities that increase their vulnerability to extreme heat. Raising the aesthetic character of the park will make it more interesting to look at for everyone, and safety improvements for one group stand to make everyone feel safer.

Any planner, designer, or urban stakeholder who wants to see their parks used and enjoyed by people of all demographic backgrounds should consider incorporating, to the fullest extent possible, these recommendations into the parks they design. By designing better parks for some, we design better parks for everyone.

References

- Alvaro-Meca, A., Palomares-Sancho, I., Diaz, A., Resino, R., De Miguel, A. G., & Resino, S. (2015). Pneumocystis pneumonia in HIV-positive patients in Spain: epidemiology and environmental risk factors. *Journal of the International AIDS Society*, 18(1), 19906.
- Anderson, E. K., & Behringer, A. (2010). Girlhood in the girl scouts. *Girlhood Studies*, 3(2), 89-108.
- Arneil, B. (2010). Gender, Diversity, and Organizational Change: The Boy Scouts vs. Girl Scouts of America. *Perspectives on Politics*, 8(1), 53–68. doi:1
- Baker, L. A., Brazel, A. J., Selover, N., Martin, C., McIntyre, N., Steiner, F. R., Nelson, A., & Musacchio, L. (2002). Urbanization and warming of Phoenix (Arizona, USA): Impacts, feedbacks, and mitigation. *Urban Ecosystems*, 6(3), 183–203.
<https://doi.org/10.1023/A:1026101528700>
- Bornioli, A., Hopkins-Doyle, A., Fasoli, F., Faccenda, G., Subiza-Pérez, M., Ratcliffe, E., & Beyazit, E. (2024). *Sex and the city park: The role of gender and sex in psychological restoration in urban greenspaces*. *Journal of Environmental Psychology*, 100, 102476.
<https://doi.org/10.1016/j.jenvp.2024.102476>
- Brown, V., Vanos, J., Kenny, N., & Lenzholzer, S. (2015). Designing urban parks that ameliorate the effects of climate change. *Landscape and Urban Planning*, 138, 118–131.
<https://doi.org/10.1016/j.landurbplan.2015.02.006>

7. References

- Bustamante, A. V., Guzman, V., Kobayashi, L. C., & Finlay, J. (2022). Mental health and well-being in times of COVID-19: A mixed-methods study of the role of neighborhood parks, outdoor spaces, and nature among US older adults. *Health & Place*, 76, 102813. <https://doi.org/10.1016/j.healthplace.2022.102813>
- Carleton College. (n.d.). *About PAR*. Participatory Action Research. <https://participatoryactionresearch.sites.carleton.edu/about-par/>
- City of Phoenix. (n.d.). *Public art map*. Phoenix Office of Arts and Culture. <https://www.phoenix.gov/arts/publicartmap>
- Cohen, D. A., McKenzie, T. L., Sehgal, A., Williamson, S., Golinelli, D., & Lurie, N. (2007). Contribution of public parks to physical activity. *American journal of public health*, 97(3), 509–514. <https://doi.org/10.2105/AJPH.2005.072447>
- Cohen, D. A., Han, B., Nagel, C. J., Harnik, P., McKenzie, T. L., Evenson, K. R., Marsh, T., Williamson, S., Vaughan, C., & Katta, S. (2016). The first national study of neighborhood parks: Implications for physical activity. *American Journal of Preventive Medicine*, 51(4), 419–426. <https://doi.org/10.1016/j.amepre.2016.03.021>
- Cokinos, C. (2024). *How to win the fight against light pollution*. Astronomy Magazine. <https://www.astronomy.com/science/how-to-win-fight-against-light-pollution/>
- Colter, K. R., Middel, A. C., & Martin, C. A. (2019). Effects of natural and artificial shade on human thermal comfort in residential neighborhood parks of Phoenix, Arizona, USA. *Urban Forestry & Urban Greening*, 39, 85–94. <https://doi.org/10.1016/j.ufug.2018.11.005>

7. References

- Condon, S., Lieber, M., & Maillochon, F. (2007). Feeling unsafe in public places: Understanding women's fears. *Revue française de sociologie*, 48(5), 101-128.
- Criado-Perez, C. (2019). *Invisible Women*. [United States], Abrams Press.
- Das, K. V., Fan, Y., & French, S. A. (2017). Park-Use Behavior and Perceptions by Race, Hispanic Origin, and Immigrant Status in Minneapolis, MN: Implications on Park Strategies for Addressing Health Disparities. *Journal of immigrant and minority health*, 19(2), 318–327. <https://doi.org/10.1007/s10903-015-0339-1>
- Declet-Barreto, J., Brazel, A. J., Martin, C. A., & others. (2013). Creating the park cool island in an inner-city neighborhood: Heat mitigation strategy for Phoenix, AZ. *Urban Ecosystems*, 16(4), 617–635. <https://doi.org/10.1007/s11252-012-0278-8>
- Denton, F. (2002). Climate change vulnerability, impacts, and adaptation: Why does gender matter? *Gender and Development*, 10(2), 10–20.
<https://doi.org/10.1080/13552070215903>
- Derosé, K. P., Han, B., Williamson, S., Cohen, D. A., & RAND Corporation (2015). Racial-Ethnic Variation in Park Use and Physical Activity in the City of Los Angeles. *Journal of urban health : bulletin of the New York Academy of Medicine*, 92(6), 1011–1023. <https://doi.org/10.1007/s11524-015-9994-8>
- Díaz, J., López, I. A., Carmona, R., Mirón, I. J., Luna, M. Y., & Linares, C. (2018). Short-term effect of heat waves on hospital admissions in Madrid: Analysis by gender and comparison with previous findings. *Environmental Pollution*, 243(Pt B), 1648–1656.
<https://doi.org/10.1016/j.envpol.2018.09.098>

7. References

- Dwyer, Allison, K. R., Goldenberg, E. R., Fein, A. J., Yoshida, K. K., & Boutilier, M. A. (2006). Adolescent Girls' Perceived Barriers to Participation in Physical Activity. *Adolescence*, 41(161), 75–89.
- Etz, K. E., & Arroyo, J. A. (2015). Small sample research: Considerations beyond statistical power. *Prevention Science*, 16, 1033-1036.
- Evans, J., Evans, S. Z., Morgan, J. D., Snyder, J. A., & Abderhalden, F. P. (2018). Evaluating the quality of mid-sized city parks: A replication and extension of the Public Space Index. *Journal of Urban Design*, 23(4), page range.
<https://doi.org/10.1080/13574809.2017.1411185>
- Evenson, K. R., Williamson, S., Han, B., McKenzie, T. L., & Cohen, D. A. (2019). United States' neighborhood park use and physical activity over two years: The National Study of Neighborhood Parks. *Preventive medicine*, 123, 117-122.
- Extreme Weather Watch. (2025). *Yearly days of 90+ degrees in Phoenix*. Extreme Weather Watch. <https://www.extremeweatherwatch.com/cities/phoenix/yearly-days-of-90-degrees>
- Folkerts, P., Bröde, P., Botzen, W. J. W., Martinius, M. L., Gerrett, N., Harmsen, C. N., Daanen, H. A., & UU LEG Research UUSE Multidisciplinary Economics. (2022). Sex differences in temperature-related all-cause mortality in the Netherlands. *International Archives of Occupational and Environmental Health*, 95(1), 249–258.
<https://doi.org/10.1007/s00420-021-01721-y>
- Fortune, D., Lord, J., Walker, E., & Froehlich, S. (2021). Enhancing belonging within

7. References

- community leisure settings. *Loisir et Société / Society and Leisure*, 44(2), 218–232.
<https://doi.org/10.1080/07053436.2021.1935420>
- Gibson, M., Loukaitou-Sideris, A., & Mukhija, V. (2019). Ensuring park equity: A California case study. *Journal of Urban Design*, 24(3), 385–405.
<https://doi.org/10.1080/13574809.2018.1497927>
- Goldsmith, L., & Bell, M. L. (2021). Queering environmental justice: unequal environmental health burden on the LGBTQ+ community. *American Journal of Public Health*, 112(1), 79-87.
- Greenfield, A. M., Alba, B. K., Giersch, G. E. W., & Seeley, A. D. (2023). Sex differences in thermal sensitivity and perception: Implications for behavioral and autonomic thermoregulation. *Physiology & Behavior*, 263, 114126.
<https://doi.org/10.1016/j.physbeh.2023.114126>
- Her City. (n.d.). Her Cities. United Nations Human Settlements Programme.
- Herman, J. L. (2013). *Gendered restrooms and minority stress: The public regulation of gender and its impact on transgender people's lives* (Research Report). The Williams Institute, UCLA School of Law.
<https://williamsinstitute.law.ucla.edu/wp-content/uploads/Restrooms-Minority-Stress-Jun-2013.pdf>
- Hernández, D., Shepard, M., & Sarnat, S. (2022). *Centering equity to address extreme heat: The role of social vulnerability in urban heat islands*. Urban Institute.
https://www.urban.org/sites/default/files/2022-02/centering-equity-to-address-extreme-heat_1.pdf

- Hewer, M., Scott, D., & Fenech, A. (2016). Seasonal weather sensitivity, temperature thresholds, and climate change impacts for park visitation. *Tourism Geographies*, 18(3), 297–321.
<https://doi.org/10.1080/14616688.2016.1172662>
- Hjort, M., & Larsen, S. H. (2025). Motives for and barriers to the use of public space among teenage girls: a scoping review. *Leisure Studies*, 1–18.
<https://doi.org/10.1080/02614367.2025.2451290>
- Hoebeke, R. (2008). Low-income women’s perceived barriers to physical activity: focus group results. *Applied Nursing Research*, 21(2), 60–65.
<https://doi.org/10.1016/j.apnr.2006.06.002>
- Höweler + Yoon Architecture. (2021). *Teen Girls Need Better Public Spaces to Hang Out* [Photograph]. Bloomberg City Lab.
<https://www.bloomberg.com/news/features/2021-05-28/we-need-more-public-space-for-teen-girls>
- Jamme, H. T. W., Bahl, D., & Banerjee, T. (2018). Between “broken windows” and the “eyes on the street:” Walking to school in inner city San Diego. *Journal of Environmental Psychology*, 55, 121-138.
- Kim, S., & Coseo, P. (2018). Urban park systems to support sustainability: The role of urban park systems in hot arid urban climates. *Forests*, 9(7), 439.
<https://doi.org/10.3390/f9070439>
- Kocmaruk, J. (2021). *Measuring park quality for youth 13-19* [Langara Applied Planning Program Major Project].

7. References

- Lane, B., Perez-Brumer, A., Parker, R., Sprong, A., & Sommer, M. (2021). Improving menstrual equity in the USA: Perspectives from trans and non-binary people assigned female at birth and health care providers. *Culture, Health & Sexuality*, 24(10), 1408–1422.
<https://doi.org/10.1080/13691058.2021.1957151>
- Lange, A. (2021). *We need more public space for teen girls*. Bloomberg.
<https://www.bloomberg.com/news/features/2021-05-28/we-need-more-public-space-for-teen-girls>
- Lewkowitz, S., & Gilliland, J. (2025). *A feminist critical analysis of public toilets and gender: A systematic review*. *Urban Affairs Review*, 61(1), 123–145.
<https://doi.org/10.1177/10780874241233529>
- Lindberg, M., & Schipperijn, J. (2015). Active use of urban park facilities – Expectations versus reality. *Urban Forestry & Urban Greening*, 14(4), 909–918.
<https://doi.org/10.1016/j.ufug.2015.08.007>
- Listerborn, C. (2016). Feminist struggle over urban safety and the politics of space. *The European Journal of Women's Studies*, 23(3), 251–264.
<https://doi.org/10.1177/1350506815616409>
- Loukaitou-Sideris, & Sideris, A. (2009). What Brings Children to the Park? Analysis and Measurement of the Variables Affecting Children's Use of Parks. *Journal of the American Planning Association*, 76(1), 89–107. <https://doi.org/10.1080/01944360903418338>
- Mahdiar, Z., & Dali, M. (2016). Adolescent use of urban parks and their social environment consequences. *Malaysia Journal of Society and Space*, 12(11), 123–136.

- Make Space for Girls. (n.d.a). *Our work*. <https://www.makespaceforgirls.co.uk/our-work-2>
- Make Space for Girls. (n.d.b). *Better Design Suggestions for Parks*.
<https://www.makespaceforgirls.co.uk/our-work-2>
- Marquet, O., Hipp, J. A., Alberico, C., Huang, J. H., Mazak, E., Fry, D., Lovasi, G. S., & Floyd, M. F. (2019). How Does Park Use and Physical Activity Differ between Childhood and Adolescence? A Focus on Gender and Race-Ethnicity. *Journal of urban health : bulletin of the New York Academy of Medicine*, 96(5), 692–702.
<https://doi.org/10.1007/s11524-019-00388-8>
- Mashhoodi. (2021). Feminization of surface temperature: Environmental justice and gender inequality among socioeconomic groups. *Urban Climate*, 40, 101004–.
<https://doi.org/10.1016/j.uclim.2021.101004>
- Meerow, S., & Keith, L. (2022). Planning for extreme heat: A national survey of U.S. planners. *Journal of the American Planning Association*, 88(3), 319–334.
<https://doi.org/10.1080/01944363.2021.1977682>
- Mehrotra, A. (2023). *A picture is worth a thousand words: Why cities must engage artists in climate action*. National League of Cities.
<https://www.nlc.org/article/2023/04/14/a-picture-is-worth-a-thousand-words-why-cities-must-engage-artists-in-climate-action/>
- Mehta, V. (2014). Evaluating Public Space. *Journal of Urban Design*, 19(1), 53–88.
<https://doi.org/10.1080/13574809.2013.854698>

7. References

- Mertens, L., Van Cauwenberg, J., Veitch, J., Deforche, B., & Van Dyck, D. (2019). Differences in park characteristic preferences for visitation and physical activity among adolescents: A latent class analysis. *PLOS ONE*, *14*(3), e0212920. <https://doi.org/10.1371/journal.pone.0212920>
- Metropolitan Area Planning Council. (2021). *Cool it with art: Using art to engage workers in climate change adaptation*. Metropolitan Area Planning Council.
- Middel, A., Chhetri, N., & Quay, R. (2015). Urban forestry and cool roofs: Assessment of heat mitigation strategies in Phoenix residential neighborhoods. *Urban Forestry & Urban Greening*, *14*(1), 178–186. <https://doi.org/10.1016/j.ufug.2014.09.010>
- Myint, S. W., Zheng, B., Talen, E., Fan, C., Kaplan, S., Middel, A., Smith, M., Huang, H.-P., & Brazel, A. (2015). Does the spatial arrangement of urban landscape matter? Examples of urban warming and cooling in Phoenix and Las Vegas. *Ecosystem Health and Sustainability*, *1*(4), 1–15. <https://doi.org/10.1890/EHS14-0028.1>
- National Aeronautics and Space Administration. (2022). *Phoenix climate: Spring 2022 analysis and technological insights* (Tech. Rep. No. 20220006753). NASA Technical Reports Server.
- Navarrete-Hernandez, V., Vetro, A., & Concha, P. (2021). Building safer public spaces: Exploring gender differences in the perception of safety in public space through urban design interventions. *Landscape and Urban Planning*, *214*, 104180. <https://doi.org/10.1016/j.landurbplan.2021.104180>
- de Nicola, F., & Giné, X. (2012). How accurate are recall data. *Evidence from coastal India*. (March 1, 2012). *World Bank Policy Research Working Paper*; (6009).

- Nilsson, A. (2017). *Frizon by Tyrens* [Photograph]. Interior Design.
<https://interiordesign.net/designwire/swedish-design-moves-new-york-spotlights-democratic-architecture/>
- Pérez-Tejera F, Valera S and Anguera MT (2018) Using Systematic Observation and Polar Coordinates Analysis to Assess Gender-Based Differences in Park Use in Barcelona. *Front. Psychol.* 9:2299. doi: 10.3389/fpsyg.2018.02299
- Peters, K., Elands, B., & Buijs, A. (2010). Social interactions in urban parks: Stimulating social cohesion? *Urban Forestry & Urban Greening*, 9(2), 93–100.
<https://doi.org/10.1016/j.ufug.2009.11.003>
- Polko, P., & Kimic, K. (2022). Gender as a factor differentiating the perceptions of safety in urban parks. *Ain Shams Engineering Journal*, 13(3), 101608.
<https://doi.org/10.1016/j.asej.2021.101608>
- Powers, S. L., Mowen, A. J., & Rodgers, E. B. D. (2024). Belonging and Welcomeness in State and Community Parks: Visitation Impacts and Strategies for Advancing Environmental Justice. *Geoforum*, 157, 104149.
- Pyry, N. (2016). Participation by being: Teenage girls' hanging out at the shopping mall as 'dwelling with' [the world]. *Emotion, Space and Society*, 18, 9–16.
<https://doi.org/10.1016/j.emospa.2016.01.006>
- Rainson, L. (2024). *Why are summers getting hotter and longer?* 12News.
<https://www.12news.com/article/weather/why-are-summer-getting-hotter-and-longer-weather-explained/75-0342ed88-ea2a-4dae-9e8b-890bf87e513e>

- Reis, H., Hino, A. A. F., Florindo, A. A., Añez, C. R. R., & Domingues, M. R. (2009). Association between physical activity in parks and perceived environment: A study with adolescents. *Journal of Physical Activity & Health*, 6(4), 503–509.
<https://doi.org/10.1123/jpah.6.4.503>
- Ries, A., Gittelsohn, J., Voorhees, C. C., Roche, K. M., Clifton, K. J., & Astone, N. M. (2008). The environment and urban adolescents' use of recreational facilities for physical activity: A qualitative study. *American Journal of Health Promotion*, 23(1), 43–50.
<https://doi.org/10.4278/ajhp.07043042>
- Rivera, E., Timperio, A., Loh, V. H., Deforche, B., & Veitch, J. (2022). Adolescents' perceptions of park characteristics that discourage park visitation. *Urban Forestry & Urban Greening*, 74, 127669.
- Roemmich, J. N., & Johnson, L. (2014). Seasonal alterations in park visitation, amenity use, and physical activity--Grand Forks, North Dakota, 2012-2013. *Preventing chronic disease*, 11, E155. <https://doi.org/10.5888/pcd11.140175>
- Schumacher, R. (2019). *Rebecca Blume Rothman, Tempe Public Art Director walks in the labyrinth at Hudson Park on Oct. 1, 2019 in Tempe, Ariz.* [Photograph]. The Arizona Republic.<https://www.azcentral.com/story/entertainment/events/2019/10/03/arizona-labyrinth-tempe-hudson-park/3777052002/>

- Seeland, Dübendorfer, S., & Hansmann, R. (2009). Making friends in Zurich's urban forests and parks: The role of public green space for social inclusion of youths from different cultures. *Forest Policy and Economics*, 11(1), 10–17.
<https://doi.org/10.1016/j.forpol.2008.07.005>
- Shaikly, A., & Mella Lira, B. (2022). Gender-bias in the built environment: how young women's experiences in public space are understood and addressed in London. *Journal of Urban Design*, 28(4), 355–374.
<https://doi-org.ezproxy1.lib.asu.edu/10.1080/13574809.2022.2125370>
- Shannon, B. (2019). Urban design and public art on Instagram. In T. Banerjee & A. Loukaitou-Sideris, A. (Eds.), *The new companion to urban design* (pp. 825–838). Routledge. <https://doi.org/10.4324/9780203731932-66>
- Smyth, E. (2016). *Arts and cultural participation among children and young people: Insights from the Growing Up in Ireland study*. The Arts Council.
<https://www.artscouncil.ie/uploadedFiles/Arts-and-cultural-participation-GUI.pdf>
- Smyth, E. (2020). *Arts & cultural participation among 17-year-olds*. The Arts Council.
https://www.artscouncil.ie/uploadedFiles/wwwartscouncilie/Content/Publications/Research_reports/Arts_and_Cultural_Participation_2020.pdf
- Strauss, R. S., Rodzilsky, D., Burack, G., & Colin, M. (2001). Psychosocial correlates of physical activity in healthy children. *Archives of pediatrics & adolescent medicine*, 155(8), 897–902. <https://doi.org/10.1001/archpedi.155.8.897>
- Stummvoll. (2004). Design against crime in vienna: A feminist approach. *Crime Prevention and Community Safety*, 6(4), 71–82. <https://doi.org/10.1057/palgrave.cpcs.8140202>

- Telford, R. M., Telford, R. D., Olive, L. S., Cochrane, T., & Davey, R. (2016). Why Are Girls Less Physically Active than Boys? Findings from the LOOK Longitudinal Study. *PloS one*, 11(3), e0150041. <https://doi.org/10.1371/journal.pone.0150041>
- Trego, S., Meerow, S., Keith, L., Schmidt, E., Leyba, B., & Jordan, L. (2023). *Plan evaluation for heat resilience: City of Phoenix, AZ*. Southwest Urban Corridor Integrated Field Laboratory (SW-IFL) – Arizona State University and University of Arizona. <https://repository.arizona.edu/handle/10150/670044>
- UN-Habitat. (2021, August 27). *Her City: Toolbox for participatory public space planning and design*. <https://unhabitat.org/news/27-aug-2021/her-city-toolbox-for-participatory-public-space-planning-and-design>
- Valentine, G. (1989). The geography of women's fear. *Area*, 21(4), 385–390. <https://doi.org/10.1111/j.1475-4762.1989.tb00933.x>
- Vert, C., Sánchez-Benavides, G., Martínez, D., Gotsens, X., Gramunt, N., Cirach, M., ... & Gascon, M. (2017). Effect of long-term exposure to air pollution on anxiety and depression in adults: a cross-sectional study. *International journal of hygiene and environmental health*, 220(6), 1074-1080.
- Veteran's Oasis Park (n.d.). Veteran's Oasis Park Map. Retrieved from <https://veteransoasispark.com>.
- Walker, S., & Clark, I. (2023). *Make space for girls: The research background 2023*. Make Space for Girls. <https://www.makespaceforgirls.co.uk/resources/research-report-2023>

7. References

Yin, S., Shen, Z., Zhou, P., Zou, X., Che, S., & Wang, W. (2011). Quantifying air pollution attenuation within urban parks: An experimental approach in Shanghai, China.

Environmental pollution, 159(8-9), 2155-2163.

Yorkshire Sport Foundation. (2022). *Make space for us*. [PDF document]. Retrieved from

https://www.yorkshiresport.org/application/files/7817/1224/5370/Make-Space-for-Us_small.pdf

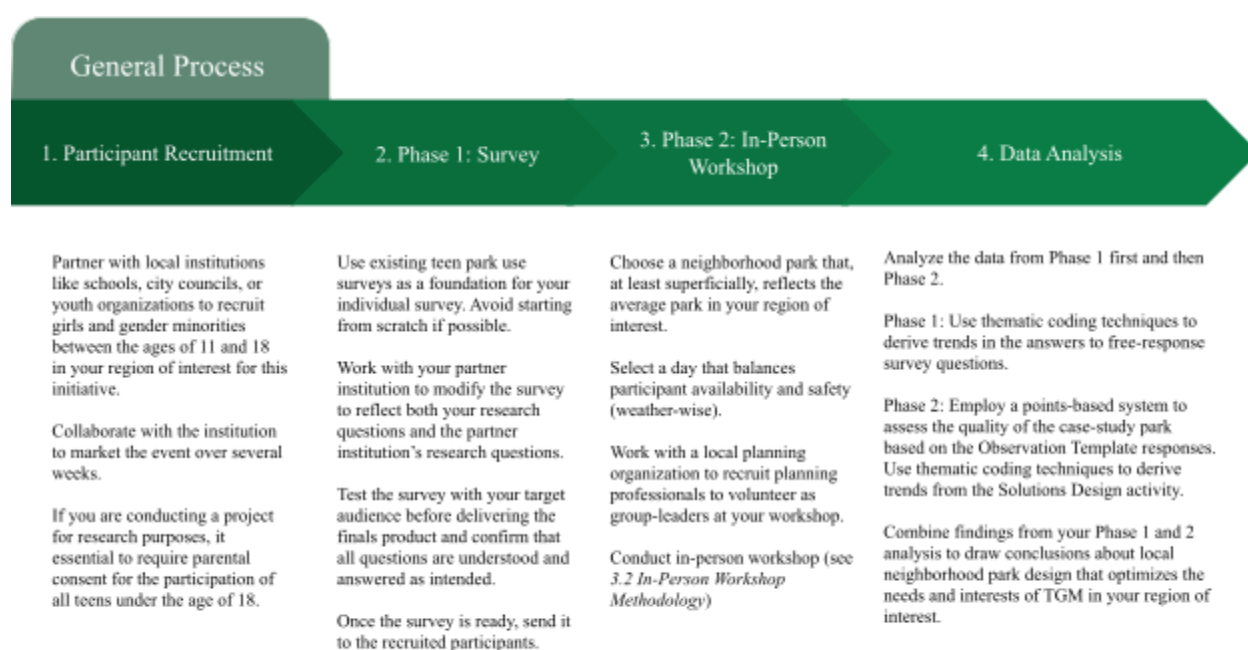
Zhang, M., Mavoa, S., Zhao, J., Raphael, D., & Smith, M. (2020). The association between green space and adolescents' mental well-being: A systematic review. *International Journal of*

Environmental Research and Public Health, 17(18), 6640.

<https://doi.org/10.3390/ijerph17186640>

Appendices

Appendix A



Appendix A. A visual depiction of the generalized research methodology.

Appendix B

**Welcome!**

Welcome to the start of your Citizen Science Journey! Completing this survey will help us gain a sense of your prior and current relationship with your neighborhood park. We encourage you to answer the questions as completely and honestly as possible. Your responses will be completely anonymous! No sensitive information will be connected to you. Please reach out to jhitzema@asu.edu or meagan.ehlenz@asu.edu if you have any questions!

Demographic Information

How old are you?

What is your gender identity?

Female

Male

Non-binary / third gender

Other (please specify)

Prefer not to say

Do you identify as a member of the LGBTQIA+ community?
(This question is optional)

Yes, I identify.

No, I do not identify.

Not Sure

Which county do you live in?

Apache
Cochise
Coconino
Gila
Graham
Greenlee
La Paz
Maricopa
Mohave
Navajo

Do you have a favorite neighborhood park?

Yes

No

What is the name and location of the park (city)?

What do you like about your favorite park? (Select all the apply)

It is close to my house/school

I feel safe there

There are places to hang out with / talk to friends

It is a good place to exercise

There are lots of trees/plants/nature

There is/are a lot of Art/Murals

It has food trucks/restaurants nearby

Other / Additional Comments

Child Park Use

When you were a kid...How often did you visit neighborhood parks?

Frequently (Multiple times per week)

Occasionally (A few times per month)

Rarely (A few times per year)

Never or Almost Never (Less than once per year)

Not Sure

Who did you visit the park with? (Select all that apply)

Parent(s)

Sibling(s)/Friend(s)

Babysitter or Nanny

Other (please specify)

What did you do at the park? (Select all that apply)

Use the playground

Play sports (organized or with friends)

Play with friends/siblings

Walk dog / play with pets

Swim/Use water features

Attend classes/community events

Attend family gatherings/picnics

Other(s)

Play games (i.e. tag, hide-and-seek, etc...)

Now, as a teenager (11+)...How often do you visit the park?

Frequently (Multiple times per week)

Occasionally (A few times per month)

Rarely (A few times per year)

Never or Almost Never (Less than once per year)

Teenage Park Use

What do you do at the park? (Select all that apply)

Exercise

Use playground facilities

Spend time with friends

Attend community events

Use sport facilities (basketball court/tennis court/skatepark)

Other

Family activities

Which features do your neighborhood parks have? (Select all that apply)

Playground

Picnic Tables

Close to my school

Swings

Shade

Garden

Basketball Court

Clean Bathrooms

Walk/Bike Path

Tennis Court

Comfortable places to sit and talk with friends

Community Events/Classes

Volleyball Court

Field

Good Lighting

Skate Park

In my neighborhood

Other(s) (please specify)

Pool/Water Feature

How frequently do you use the following park features? (If a park feature you use is not listed, please write it in the "Other" rows)

	Never	Rarely	Often	Always	Does not apply
Playground	☐	☐	☐	☐	☐
Swings	☐	☐	☐	☐	☐
Basketball Court	☐	☐	☐	☐	☐

8. Appendices

Tennis Court	☐	☐	☐	☐	☐
Volleyball Court	☐	☐	☐	☐	☐
Skate Park	☐	☐	☐	☐	☐
Pool/Water Feature	☐	☐	☐	☐	☐
Picnic Tables	☐	☐	☐	☐	☐
Bathrooms	☐	☐	☐	☐	☐
Field	☐	☐	☐	☐	☐
Garden	☐	☐	☐	☐	☐
Walk/Bike Path	☐	☐	☐	☐	☐
Other 	☐	☐	☐	☐	☐
Other 	☐	☐	☐	☐	☐
Other 	☐	☐	☐	☐	☐

Are there parks near you or parts of your local park that you don't like or you try to avoid?
Why?

Why do you think you don't go to the park?

Why do you think you stopped going to the park?

Why don't you go to the park as often as when you were younger?

If you could make any changes to your neighborhood park, what would they be?

Can you think of anything that would make you more likely to visit the park?

Climate

Please rank the following seasons in order of when you are most likely to visit a neighborhood park?

Spring

Summer

Fall

Winter

How does heat impact your decision to visit the park?

I am less likely to use the park on hot days.

The temperature does not affect whether or not I visit the park.

I am more likely to use the park on hot days.

What time of the day are you most likely to visit the park in the summer?

Morning

Afternoon

Evening

Night

I would go to the park more in the summer if...

	Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree
...I lived somewhere with cooler summer	☐	☐	☐	☐	☐

temperatures.

...there was more shade. ☐ ☐ ☐ ☐ ☐

...there was a water feature(s) like a pool or splashpad. ☐ ☐ ☐ ☐ ☐

...I felt comfortable going at night. ☐ ☐ ☐ ☐ ☐

...there were indoor facilities to help me escape the heat. ☐ ☐ ☐ ☐ ☐

Park Perceptions

When I go to my neighborhood park...

	Strongly Disagree	Somewhat Disagree	Neither agree nor disagree	Somewhat Agree	Strongly Agree
I feel comfortable being there alone.	☐	☐	☐	☐	☐
I go/hang out with friends of my same gender.	☐	☐	☐	☐	☐
I feel comfortable being there at night.	☐	☐	☐	☐	☐
I generally feel safe.	☐	☐	☐	☐	☐
There is plenty for me to do.	☐	☐	☐	☐	☐
There are plenty of things for teenagers to do.	☐	☐	☐	☐	☐
There are plenty of things for teenage girls and non-binary people to do.	☐	☐	☐	☐	☐

8. Appendices

There are plenty of things for teenage boys to do.	☐	☐	☐	☐	☐
I feel it is more for little kids than it is for me.	☐	☐	☐	☐	☐
I feel like I belong.	☐	☐	☐	☐	☐
The restrooms are clean.	☐	☐	☐	☐	☐
The restrooms provide me with enough privacy.	☐	☐	☐	☐	☐
Sometimes I feel intimidated by groups of other teens.	☐	☐	☐	☐	☐

My parents...

	Strongly Disagree	Somewhat Disagree	Neither agree nor disagree	Somewhat Agree	Strongly Agree
Encourage me to spend time at the park.	☐	☐	☐	☐	☐
Are comfortable with me going to park alone.	☐	☐	☐	☐	☐
Prefer I do not go to the park.	☐	☐	☐	☐	☐

Powered by Qualtrics

Appendix C

Preliminary Observations

When collecting data, we want to make sure that we know certain facts about the subject to make sure our experiment is *controlled*. Record the following information before beginning your audit:

What is the name and postal/zip code of the park you are auditing?

Name: _____ Zip Code: _____

Girls Scout Grade Level (e.g. Cadette) _____

Number of Group Members _____

Temperature (°F) _____

% Humidity _____

Date _____

Time _____

Day of the Week _____

Number of People in the Park _____

Open Hours of the Park _____

Is there a school near the Park ? **Yes** **No**

Design and Aesthetic

For park visitors to enjoy the park, it needs to look interesting and exciting. What do you observe about the attractiveness of the park?

What are your first impressions of how the park looks? What looks interesting, attractive, unsafe, etc...? Do any parts of the park that provoke certain emotions? Write your feelings in the space below.

Nature: What fraction of the park is covered in natural features like trees, grass, and plants (Circle)?

0 $\frac{1}{3}$ $\frac{1}{4}$ $\frac{2}{3}$ $\frac{1}{2}$ $\frac{2}{3}$ $\frac{3}{4}$ 1

Visual Appeal: Describe the aesthetic quality of the park? (Select all that you can find)

- | | |
|--|---|
| ☐ 1 or more murals/paintings in the park | ☐ Sidewalk or pathway art present |
| ☐ Playground equipment is colorful and attractive | ☐ Interesting seating areas |
| ☐ Would be a good place to take photos for social media | ☐ Creative/pretty landscape design |

Additional Notes:

Focal Points: Are there any parts or features of the park that are particularly eye catching? List them and take a picture of each one. Write why they catch your attention on the back.

Comfort (Safety)

A high quality park makes its visitors feel comfortable. There should be features that make the park fun and safe for everyone! Make the following observations about different aspects of the park

Safety: Which of the following safety features can you find in the park?

- ☐ Fences/gates around the park
- ☐ Locks/doors on the bathroom stalls
- ☐ Few corners or places for someone to hide
- ☐ Spaces to enter or exit the park on all sides
- ☐ The street is visible from any part of the park
- ☐ Lighting to keep the park visible and safe at night
- ☐ Crosswalks, sidewalks, and/or paths that allow someone to enter/exit/walk around the park safely at all times

Additional Comments about Park Comfort:

On a scale of 1 to 5 (1 being completely disagree and 5 being, completely agree), **how much would you (or your group) agree with the following statement?** Please explain your answer on the lines below.

If there were an emergency or you / someone in this park was in danger, they would easily be able to get help.

1 2 3 4 5

Comfort (Maintenance and Environment)

A high quality park makes its visitors feel comfortable. There should be features that make the park fun and safe for everyone! Make the following observations about different aspects of the park:

Maintenance: What observations can you make about the cleanliness and maintenance of the park?

- ☐ There is little or no litter/dog poop scattered around the park. Most garbage is properly disposed of.
- ☐ The bathrooms are clean and hygienic (i.e. soap/toilet paper available, clean toilets, okay smell, etc...).
- ☐ There is little or no offputting graffiti or vandalism.
- ☐ The park appears generally well taken care of.

Environment: Which features does the park have to help visitors deal with environmental challenges?

- ☐ Benches/places to rest all around the park
- ☐ Private places to socialize that are away from active areas like the playground or basketball courts
- ☐ Abundance of shade (Artificial – ramadas and canopies– or Natural – trees and bushes)
- ☐ Water fountains or places to easily access water
- ☐ Quiet/peaceful places to escape noise (from traffic or people) in this park

Additional Notes: How else would you describe the park's comfort, maintenance, and environment overall? Does anything else stick out to you?

Activities

Parks are places for recreation! The best parks are built to accommodate a wide range of activities and uses. Make observations about park features and activities to evaluate the park!

Locate the Park Rules: Are there any rules that make the park seem less fun or inviting?

Please list them in the box below

Range of Activities: Take a minute to look around the park...

List the features you see in the park (i.e. bathroom, pool, playground, tennis court, ramada, etc...)

Brainstorm 5 (or more) activities you could imagine **YOURSELVES** doing in the park and where (e.g. picnic on the grass, playing on the playground, game on the basketball court).

- | | |
|----------|-------------------|
| 1. _____ | 4. _____ |
| 2. _____ | 5. _____ |
| 3. _____ | Additional: _____ |

Brainstorm 5 activities you could imagine **SOMEONE ELSE** doing in the park and where. (e.g. picnic on the grass, playing on the playground, game on the basketball court). Be sure to specify what type of person you would imagine doing the activity (i.e. elderly woman, teenage boy, little kid, etc...)

- | | |
|----------|-------------------|
| 1. _____ | 4. _____ |
| 2. _____ | 5. _____ |
| 3. _____ | Additional: _____ |

Reflections

Reflect on your visit to the park to draw a conclusion about whether or not the park is inclusive of teenage girls and gender diverse people.

Would you feel safe in this park at all times of the day?

☐ Yes ☐ No

Are the amenities in this park interesting to you?

☐ Yes ☐ No

If you were at this park and you had to use the restroom, what would you do?

☐ Use the park bathroom ☐ Use the bathroom of a store or restaurant nearby
☐ Hold it ☐ Go home ☐ Other

Are there any groups of people in this park that may make you uncomfortable?

☐ Yes, specify: _____ ☐ No

If you were making plans with your friends, would you suggest visiting this park?

☐ Yes ☐ No

If you were to visit this park, what would you do here? Who would you come with?

FINAL CONCLUSION: Is this park inclusive of the needs and interests of teenage girls?

☐ Yes ☐ No

Why? Explain your conclusion in the box below.

Do you think other girls (or non-male people) your age would feel the same as you?

☐ Yes ☐ No

Appendix D



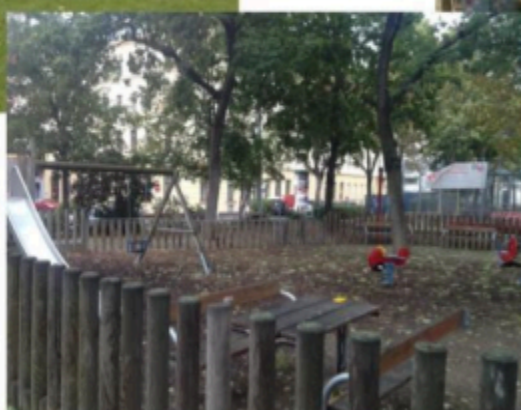
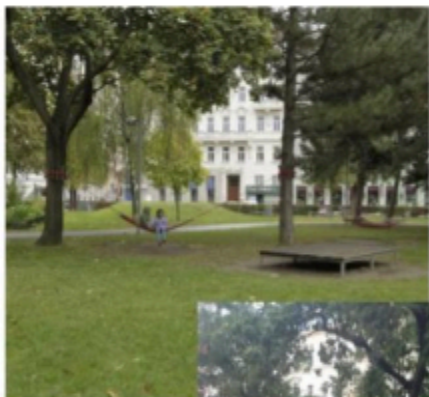
From Bloomberg CityLab interview with Susannah Walker,
Teen Girls Need Better Public Spaces to Hang Out

"When the playscape Swing Time popped up in Boston in 2014, visitors started having too much fun. The **20 hoop-shaped swings suspended from a white shade structure light up when in use, glowing purple with vigorous motion.** Its creators at Höweler + Yoon Architecture had imagined people would swing in ones and twos. Instead groups tried to pile on together, hoping to share the sway and have a conversation."

"One, girls love swings. And two, **there aren't enough swings made for teenage girls.** "They are almost always placed with the equipment for younger children, so that if teenagers use them they are seen as invaders."

"At the end of the summer holidays my friend and I ran out of money," Walker wrote in a March post. "We had nothing to do and there was nowhere to go. So **we'd go and hang out on the swings in the early evening and chat as the light slowly faded into dusk.** It was better than sitting around at home."

Appendix D.1. Swingtime inspiration image (above) with description (below)



From Make Space for Girls,

Vienna Case Study

"girls often passed through the park on their way to and from school. So the first strategy was to get them to stop for just a few minutes, described as 'quick attraction' by the planners. They **put up hammocks and platforms near the paths, to give the girls a reason to stop.** This began to work, and more girls lingered."

"Their fear of the park was also addressed with **wider paths which ran round the whole perimeter, and better lighting.**"

"The next step was to **alter the games area so that the boys could not dominate it** and the girls felt safe to go in. The **entrances were increased to three, and widened so that the girls did not feel trapped.** The playing area was also divided into two with a block which could be a seating area but also part of a game."

"Some girls came from ethnic backgrounds which meant that they were expected to care for their younger siblings, and mostly came to the park with them, so seats and benches were installed in the infants playground so that they could sit and chat while keeping an eye on their siblings."

Appendix D.2. Vienna Case Study inspiration image (above) with description (below)



Appendix D.3. Myanmar Case Study inspiration image (above) with description (below)



From Make Space for Girls,

Umea Case Study

“Safety is a big barrier to teenage girls accessing open spaces. Elsewhere in Umeå, the attention to how women experience the city resulted in a narrow and dark underpass which women and girls found unsafe being redesigned.”

“The lighting was improved and the shape of the entrances pared back with curves, resulting in better lines of sight and fewer hiding places. An additional exit was also added in the middle, so that women can get out if they are feeling uncomfortable for any reason.”

Appendix D.4. Umea Case Study inspiration image (above) with description (below)



From Make Space for Girls,

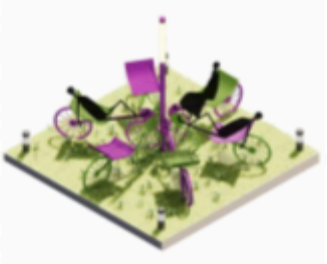
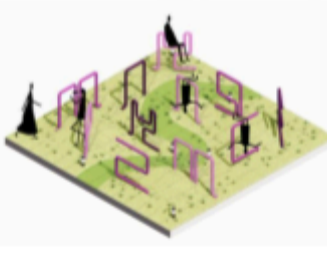
Umea Case Study: Frizon in Årstadernas Park

"Tyrens architects and the artist Kerstin Bergendal designed it in collaboration with local teenage girls over a number of workshops. The results are beautiful but also practical."

"It's social seating – what's not to like – and has shelter against Swedish winter weather. But there's more to it than that. **The seats are also designed to be ergonomically right for teenage girls, rather than for the default male.** And you can play music on it from your phone."

"The elegance is also important, it's a sign to the teenage girls that their presence is welcomed and valued in the park. And it works – research shows that it is well used by teenage girls right into the evening."

Appendix D.5. Umea Case Study: Frizon in Artsidernas Park inspiration image (above) with description (below)

Still in the gym "Outdoor gym facilities tend to be dominated by young men. But we have heard rumours of a park (in north Bristol perhaps?) where a social circle of recumbent bikes has been taken over by the teenage girls because they appreciate the chance to chat and exercise at the same time." 	Lessons from Vienna and Glasgow "They experimented with what made girls linger in Einsiedler Park, Vienna and found that hammocks worked. It turns out that the hammocks in Hayburn Park in Glasgow are also full of teenage girls." 
Hanging Around "Inspired by Eivars Rodda Matta, Malmö, the first park to be designed by and for girls age 10-14. They specifically wanted gym bars – high enough to hang from, swing round, lean against, look around, but not so tall that they become a magnet for young men wanting to do chin ups." 	Dividing a MUGA "All the research shows that if there's single pitch in a MUGA, the boys will dominate it. So dividing up so there is not just a pitch but also a less prescriptive space gives girls a better chance to play. The divider can be whatever you want: it's a goal, a shelter, a place for TikTok dances." 
Social Seating "Girls like to face each other when they chat so traditional park benches along a path means someone ends up sitting on the floor, and benches lined up to watch the action on the MUGA aren't great either. Simple picnic benches like the ones in the image above create a more social space or more creatively, seating that echoes a fairground walker..." 	For when it is too hot or too rainy "Kind & Samerding have done some fantastic research with young teens in Belgium; a group of girls were sad that "their" shelter had fallen into disrepair; they imagined some wonderful mushroom shapes to replace it, which inspired this look... and the shelters are designed to collect rainwater too." 
Performance in the Park "Back to Rhems Rodda Matta, Malmö, where the girls designed a stage, which has been great for performances (formal/informal) and used by the community for fitness classes etc; the tree log stools provide social space and double up as stepping stones." 	Up high "No lock down filled there were a few (hey-tell) months where teenagers could meet in groups of six but couldn't go into each others houses. So they had to be creative about where they could hang out. And we noticed how many could be seen at the top of climbing frames. It turns out that being high up in fun, so some (non-prescriptive) high up hang out places, with scramble net for access." 

Appendix D.6. Inspiration images from Make Space for Girls: What Does Better Look Like?

Appendix E

Map of Hudson Park used for Solutions Design activity.



Appendix E. Map of Hudson Park used for Solution Design activity.

Appendix F

Results of Statistical Tests

<i>Ranks</i>		N	Mean Rank	Sum of Ranks
Reported Park Visitation Frequency Teen vs. Child (Recalled)	Positive Ranks	0	433.5	0
	Negative Ranks	34	433.5	867
	Ties	8	36	36
	Total	42		

<i>Test Statistics</i>	
H_0 : Reported Teen Park Visitation Frequency is equal to Reported (Recalled) Child Park Visitation Frequency.	
Z-Score	-5.573
Prob > z 	0.000

Appendix F.1. Results of the Wilcoxon Signed-Rank Test comparing Reported Teenage Park Visitation Frequency with Recalled Childhood Park Visitation Frequency

<i>Ranks</i>		N	Mean Rank	Sum of Ranks
Agreement Belonging Statement	14 Years Old or Younger	24	480	539.5
	Older than 14 Years	15	300	240.5
	Total	39		

<i>Test Statistics</i>	
H_0 : Both age groups agree to the same extent	
Z-Score	1.771
Prob > z 	0.0766

Appendix F.2. Results of the Wilcoxon Rank Sum Test Comparing Reported Agreement that there is Plenty for the Respondent to Do at their Neighborhood Park by Age Group (Older than 14 Years vs. 14 years or Younger)

<i>Ranks</i>		N	Mean Rank	Sum of Ranks
Agreement Belonging Statement	14 Years Old or Younger	24	480	562
	Older than 14 Years	15	300	218
	Total	39		

<i>Test Statistics</i>	
H_0 : Both age groups agree to the same extent.	
Z-Score	2.464
Prob > z 	0.0137

Appendix F.3. Results of the Wilcoxon Rank Sum Test Comparing Reported Agreement that They Belong in their Neighborhood Park by Age Group (Older than 14 Years vs. 14 years or Younger)

<i>Ranks</i>		N	Mean Rank	Sum of Ranks
Reported Park Visitation Frequency Teen vs. Child (Recalled)	Positive Ranks	1	301.5	33.5
	Negative Ranks	17	301.5	569.5
	Ties	24	300	300
	Total	42		

<i>Test Statistics</i>	
H_0 : Reported engagement in elective park activities as a teen is equal to reported engagement in elective park activities as a child.	
Z-Score	-3.771
Prob > z 	0.0002

Appendix F.4. Results of the Wilcoxon Signed-Rank Test comparing reported engagement in elective park activities as a teenager and as a child (recalled)

Age Range	Reports Playground Use		Total	P-Value
	Yes	No		
11-12 years	7	6	13	0.837
13-14 years	4	6	10	
15-16 years	3	5	8	
17+ years	1	2	3	
Total	15	19	34	

Appendix F.5. Results of the Chi-Squared Test of Reported Playground Use (Yes vs. No) by Age Category

<i>Ranks</i>				
		N	Mean Rank	Sum of Ranks
Agreement that there is plenty for teens to do in their neighborhood park	Did not report Community Events	23	326.5	368
	Reported Community Events	8	169.5	128
	Total	31		
<i>Test Statistics</i>				
H₀: No relationship between agreement that there is plenty for teens to do in their neighborhood park and whether they reported the presence of community events in their neighborhood park.				
Z-Score		-1.929		
Prob > z 		0.0537		

Appendix F.6. Results of the Wilcoxon Rank Sum Test Comparing Reported Agreement that There is Plenty for Teens to Do in their neighborhood park and whether they report the presence of community events in their neighborhood park.

Appendix G

	Likert Percent Shares				
Belonging Statement	Strongly Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Strongly Agree
There is plenty for me to do.	15.38	23.08	7.69	25.90	17.95
There are plenty of things for teenagers to do.	30.77	28.21	12.82	23.08	5.14
There are plenty of things for teenage girls and non-binary people to do.	15.38	25.64	25.64	27.95	15.38
There are plenty of things for teenage boys to do.	10.26	25.64	20.51	25.64	17.95
I feel it is more for little kids than it is for me.	2.56	7.69	12.82	38.46	38.46
I feel like I belong.	7.69	20.51	38.46	20.51	12.82

Appendix G.1. Table displaying the percent breakdown of the extent to which respondents agree with various identity belonging statements regarding their local park.

8. Appendices

	Likert Percent Shares (N=39)				
Statement	Strongly Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Strongly Agree
I feel comfortable being there alone.	17.95	17.95	23.08	25.64	15.38
I feel comfortable being there at night.	15.38	28.21	20.51	20.51	15.38
I generally feel safe.	2.56	15.38	15.38	41.03	25.64
Sometimes I feel intimidated by groups of other teens.	12.82	12.82	23.08	28.21	23.08
The restrooms are clean.	28.21	10.26	46.15	12.82	2.56
The restrooms provide me with enough privacy.	20.51	7.69	48.72	12.82	10.26

Appendix G.2. Table displaying the percent breakdown of the extent to which respondents agree with various comfort statements regarding their local park.

Appendix H



Appendix H. Solutions map recommending that an “insta wall (mural)” be added the ramp in the skate park.